Exercises on Mathematics 2

Duong Thanh PHAM

January 31, 2026

Chapter 1

Integration

1.1 Integration techniques

Exercise 1.1.1. Evaluate the integral $\displaystyle\int \frac{x}{\sqrt{1+x^2}}\,dx$

Exercise 1.1.2. Evaluate the integrals $\displaystyle\int_{-1}^{1} (1+|t^3|)\,dt$.

Exercise 1.1.3. Evaluate

(a) $\displaystyle\int_0^1 e^x \sin(e^x)\,dx$

(b) $\displaystyle\int_1^2 x \ln x\,dx$

(c) $\displaystyle\int_0^1 x^2 e^x dx$

(d) $\displaystyle\int_1^2 \ln(3x)\,dx$

(e) $\displaystyle\int_1^2 \frac{\ln x}{x^2}\,dx$

Exercise 1.1.4. Given the the table that shows the values of f, f', g, g' at $x=0$ and $x=1$:

x	0	1
f(x)	2	4
f'(x)	6	-3
g(x)	-4	3
g'(x)	2	-1

Let $\displaystyle\int_0^1 f'(x)g(x) = 5$. Evaluate $\int_0^1 f(x)g'(x)\,dx$.

Exercise 1.1.5. Evaluate the integrals

(i) $\displaystyle\int (\sin(2x)+\cos(2x))\,dx$

(ii) $\displaystyle\int \frac{\sin x\sqrt{5-\cos x}}{2}\,dx$

(iii) $\displaystyle\int \frac{(\sin x)^3}{\cos x+2}\,dx$

(iv) $\displaystyle\int (\sin x+8)\cos^3 x\,dx$

Exercise 1.1.6. Evaluate $\displaystyle\int_0^{\pi/2} \sin x\,\cos x\sqrt{1+\sin^2 x}\,dx$.

Exercise 1.1.7. Find A and B such that

$$\frac{1}{x^2+2x-3} = \frac{A}{x-1} + \frac{B}{x+3} \quad \forall x \neq 1,\ x \neq 3.$$

Then evaluate the integral

$$\int_2^5 \frac{1}{x^2+2x-3}\,dx.$$

Exercise 1.1.8. Evaluate

(a) $\displaystyle\int_3^4 \frac{-x-12}{(x+5)(x-2)}\,dx$ (b) $\displaystyle\int_2^3 \frac{x-8}{3x^2+2x-8}\,dx.$

Exercise 1.1.9. Evaluate the following integrals

$$\int_0^{\ln 2} \frac{e^{2t}}{e^{2t}+3e^t+2}\,dt.$$

Exercise 1.1.10. Evaluate the integrals

(i) $\displaystyle\int \frac{dx}{x^2\sqrt{9-x^2}}.$ (ii) $\displaystyle\int_0^1 \frac{dx}{(x^2+1)\sqrt{x^2+1}}$

Exercise 1.1.11. Evaluate

$$\int_0^1 \sqrt{t^4+t^6}\,dt.$$

1.2 Fundamental theorems of Calculus

Exercise 1.2.1. Find a function $y = f(x)$ such that

$$\int_0^x f(t)\,dt = \frac{\sin x}{x^2+1} \quad \text{for all } x.$$

Exercise 1.2.2. Find a continuous function $f(x)$ and a number $a > 0$ such that

$$16 + \int_a^x t^2 f(t)\,dt = x^4$$

for all values of x.

(Hint: Differentiate both sides with respect to x.)

Exercise 1.2.3. Find a continuous function $f(x)$ on $(-\infty, \infty)$ such that

$$\int_0^{2017x} f(t)\,dt = \frac{e^x}{x^2+1} - 1, \quad \forall x \in (-\infty, \infty).$$

Exercise 1.2.4. Let $F(x) = \displaystyle\int_x^{2x+1} \sqrt{t^2+2t+5}\,dt$. Find $F'(x)$.

Exercise 1.2.5. Given $F(x) = \displaystyle\int_1^{\pi x} \sqrt{t+\sin t}\,dt$, find value of $F'(1)$.

Exercise 1.2.6. Let $F(x) = \displaystyle\int_{x^2}^{x^3} \sqrt{t^4+1}\,dt$. Find $F'(x)$.

Exercise 1.2.7. Evaluate

(a) $\lim_{x\to 0} \dfrac{\int_0^x \cos(t^2)dt}{x}$ (b) $\lim_{x\to 0} \dfrac{\int_0^x \cos(\pi t^2)dt}{x}$ (c) $\lim_{x\to 0} \dfrac{\int_{x^2}^0 \sqrt{1+t^3}dt}{x^2}$.

Exercise 1.2.8. Let $f'(x) = 2x - 3\sqrt{x}$ and $f(0) = 3$. Find $f(1)$.

Exercise 1.2.9. Given $f''(x) = 12x^2 + e^x$, $f'(0) = 4$ and $f(0) = 1$, find $f(x)$.

Exercise 1.2.10. An oil storage tank cracks at time $t = 0$ and oil leaks from the tank at a rate of $r(t) = e^{-0.1t}$ liters per minute. How much oil leaks out during the first ten minutes?

Exercise 1.2.11. An electronics factory uses a certain rare metal. The metal is used up at a rate $r(t) = 2e^{0.04t}$ tons per year, where t is time in years. At time $t = 0$ the factory has a reserve of 40 tons. How long will it take for the factory to use up this reserve?

Exercise 1.2.12. A particle moves along a straight line and its velocity at time t is given by $v(t) = 2t + \cos 2t$. If its initial position is $s(0) = 1$, then its position at $t = T$ is

Exercise 1.2.13. Assume that $f(1) = 10$ and $f'(x) \geq 2$ for all values of $x \in [1, 4]$. Which of the following statements must be true?

(A) $f(4) < 25$

(B) $f(4) \geq 16$

(C) $f(4) \leq 18$

(D) $f(4) \leq 12$

Exercise 1.2.14. Use the Midpoint Rule by dividing the interval $[0, 1]$ into 5 equal subintervals to evaluate the integral

$$\int_0^1 \sqrt{1+x^3}\, dx.$$

Exercise 1.2.15. By using the Simpson rule and dividing the interval $[0, 1]$ into 6 equal subintervals, approximate the integral

$$\int_0^3 \sqrt{1+x^3}\, dx.$$

(Give your answer correct to at least 3 decimal places.)
How many subintervals should we use if we want to guarantee that the error will be less than 10^{-4}?

Exercise 1.2.16. For the function f shown in the figure, the approximate value of the integral $\int_0^6 f(x)\, dx$ found using six rectangles of equal width and right hand endpoints is

(A) 19 (B) 20 (C) 21 (D) 23.

1.3 Improper integrals

Exercise 1.3.1. Evaluate

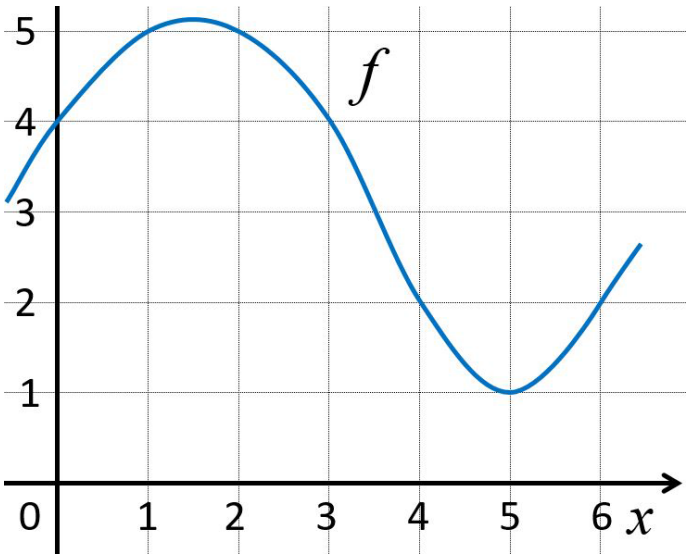


Figure 1.1: This figure is used in Exercise 1.2.16

(a) $\displaystyle\int_0^\infty (x+1)e^{-2x}\,dx$ (c) $\displaystyle\int_0^\infty (2x-1)e^{-x}\,dx$ (d) $\displaystyle\int_e^{2e} \frac{1}{x\sqrt{\ln x - 1}}\,dx$ (e) $\displaystyle\int_1^\infty \frac{dx}{x^5}$

(b) $\displaystyle\int_0^\infty xe^{-x^2+1}\,dx$

Exercise 1.3.2. Find the area under the curve $y = \dfrac{1}{(x+2)^2}$ from $x = 0$ to $x = \infty$.

Exercise 1.3.3. Show that the following improper integral converges

$$\int_0^\infty \frac{\arctan x}{(1+x^2)^{2017}}\,dx.$$

Exercise 1.3.4. Evaluate the integral $\displaystyle\int_{-1}^2 \frac{8}{x^3}\,dx$.

Exercise 1.3.5. In probability theory, the mean of the standard normal distribution is

$$\mu = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty x\, e^{-x^2/2}\,dx.$$

Evaluate μ.

Exercise 1.3.6. Let R be the region between the graph of $f(x) = e^{-2x}$ and the x-axis for $x \geq 3$. Find the area of R.

Exercise 1.3.7. The value of $\displaystyle\int_1^2 \frac{dx}{\sqrt{x-1}}$ is

(A) ∞ (B) undefined (C) $\dfrac{1}{2}$ (D) 2

Chapter 2

Applications of Integration

2.1 Average value of a function, Areas, volumes, arc lengths

Exercise 2.1.1. The average value of the function $f(x) = \dfrac{1}{1+x^2}$ on $[0,1]$ is

(A) $\dfrac{1}{2}$ (B) $\dfrac{3}{4}$ (C) $\dfrac{1}{2}\ln 2$ (D) $\dfrac{\pi}{4}$

Exercise 2.1.2. A cup of hot tea with initial temperature 90° C is placed in a room where the temperature is 25° C. According to Newton's Law of Cooling, the temperature of the tea after t minutes is

$$T(t) = 25 + 65e^{-\frac{t}{50}}.$$

Evaluate the average temperature of the tea during the first hour.

(Hint: The average value of a continuous function f on $[a,b]$ is defined to be the number $\frac{1}{b-a}\int_a^b f(x)dx$.)

Exercise 2.1.3. Consider the region A bounded by the curves $y = x^2 - 1$ and the line $y = 3x - 3$.

(a) Find the area of A.

(b) Find volume of the solid obtained by rotating A about the x-axis.

Exercise 2.1.4. Let G be the region enclosed by the the curves $y = x^{3/2}$, $x = 0$, $x = 4$ and $y = 0$.

(a) Find the area of of the region G.

(b) Find the volume of the solid generated by rotating the region G about the x-axis.

(c) Find the circumference of the region G.

Exercise 2.1.5. A torus (a donut-shaped solid) is generated by rotating a circle of radius 2 cm about an axis 5 cm from the center of the circle, as shown in the figures below (Figure 2.1). Find the volume of the torus.

[Hint: Use the method of cylindrical shells, $V = \displaystyle\int_a^b 2\pi x f(x)\,dx$].

Exercise 2.1.6. Find the area of the region enclosed by $y = x^2 - 2x$ and $y = x + 4$.

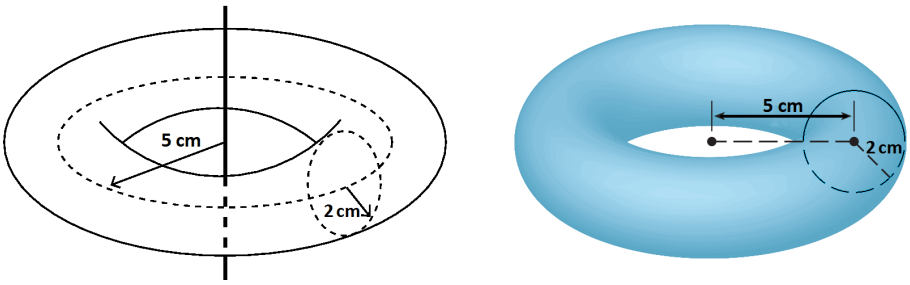


Figure 2.1: This figure is used in Exercise 2.1.5

Exercise 2.1.7. Find the area of the region enclosed by $y = \sqrt{x^3}$ and $y = x$.

Exercise 2.1.8. Determine the area of the region bounded by $y = 4x + 3,\ y = 6 - x - 2x^2$.

Exercise 2.1.9. Consider the region R bounded by the curves $y = 5 - x^2$ and $y = 4x^2$.

(i) Find the area of R.

(ii) Find volume of the solid obtained by rotating R about the x-axis.

Exercise 2.1.10. Let R be the region enclosed by $y = 12 - x^2$ and $y = |x|$.

(i) Sketch the region R.

(ii) Find the area of R.

Exercise 2.1.11. Let G be the region enclosed by the curves $y = \frac{4}{x}$ and $y = 5 - x$

(i) Find the area of the region G.

(ii) Find the volume of the solid generated by rotating the region G about the x-axis.

Exercise 2.1.12. Find the area of the region S enclosed by $y = x$ and $y = 5x - x^2$.

Exercise 2.1.13. A city's water storage tank is in the shape of the solid of revolution generated by rotating the region under the graph of

$$f(x) = 20\sqrt{1 - \frac{x^2}{30^2}}$$

from $x = -30$ to $x = 30$m about the x-axis. What is the volume of this tank?

Exercise 2.1.14. The region bounded by the curves $y = \sqrt{x},\ x = 0$ and $y = 1$ is revolved around the x-axis. The volume of the solid generated is given by which integral?

(A) $\pi \int_0^1 x\,dx$

(B) $\pi \int_0^1 (1 - x)\,dx$

(C) $\pi \int_0^1 x(1 - \sqrt{x})\,dx$

(D) $\pi \int_0^1 (1 - \sqrt{x})^2\,dx$

Exercise 2.1.15. Which integral represents the volume of the solid generated when the region under the curve $y = \cos x$ on $[0, \pi/2]$ is rotated about the y-axis?

(A) $\pi \int_0^{\pi/2} \cos x \, dx$

(B) $2\pi \int_0^1 \cos x \, dx$

(C) $2\pi \int_0^{\pi/2} x \cos x \, dx$

(D) $\pi \int_0^{\pi} \sin^2 x \, dx$

2.2 Others

Exercise 2.2.1. Let f be a continuous and decreasing function on $[0, 1]$. Show that

$$\int_0^{1/2} f(x)dx \geq \frac{1}{2} \int_0^1 f(x)dx.$$

Exercise 2.2.2. The demand function for a certain type of seed is

$$p(x) = \frac{500}{(3x+4)^2}.$$

Find the consumer surplus when the production level x is 2 units.
(Hint: The demand function is $p(x)$, where p is price and x is production level. The consumer surplus at production level X is defined as $\int_0^X [p(x) - P]dx$ where $P = p(X)$.)

Exercise 2.2.3. (a) Show that

$$1 \leq \sqrt{1+x^3} \leq 1 + x^3 \quad \text{for all } 0 \leq x \leq 1.$$

(b) Show that

$$1 \leq \int_0^1 \sqrt{1+x^3} \, dx \leq \frac{5}{4}.$$

Exercise 2.2.4. The expression

$$\frac{1}{50}\left(\sqrt{\frac{1}{50}} + \sqrt{\frac{2}{50}} + \cdots + \sqrt{\frac{50}{50}}\right)$$

is the Riemann sum approximation for

(A) $\int_0^1 \sqrt{\frac{x}{50}} \, dx$

(B) $\int_0^1 \sqrt{x} \, dx$

(C) $\frac{1}{50} \int_0^1 \sqrt{\frac{x}{50}} \, dx$

(D) $\frac{1}{50} \int_0^1 \sqrt{x} \, dx$

Exercise 2.2.5. If $f(1) = 3$ and $\displaystyle\int_0^1 f(x)dx = 2$, then the value of $\displaystyle\int_0^1 xf'(x)dx$ is

(A) 3 (B) 2 (C) 5 (D) 1

Chapter 3

Differential Equations

Exercise 3.0.1. Show that

$$y = \frac{3}{2}e^x + e^{-2x}$$

is a solution of the differential equation

$$y' + 2y = 2e^x.$$

Exercise 3.0.2. Verify that

$$y = -t\cos t - t$$

is a solution of the initial-value problem

$$t\frac{dy}{dt} = y + t^2 \sin t, \qquad y(\pi) = 0.$$

Exercise 3.0.3. (a) For what values of r does the function $y = e^{rx}$ satisfy the differential equation

$$2y'' + y' - y = 0?$$

(b) If r_1 and r_2 are the values of r that you found in part (a), show that every member of the family of functions

$$y = ae^{r_1 x} + be^{r_2 x}$$

is also a solution.

Exercise 3.0.4. (a) For what values of k does the function $y = \cos kt$ satisfy the differential equation

$$4y'' = -25y?$$

(b) For those values of k, verify that every member of the family of functions

$$y = A\sin kt + B\cos kt$$

is also a solution.

Exercise 3.0.5. Which of the following functions are solutions of the differential equation

$$y'' + y = \sin x?$$

(a) $y = \sin x$ (b) $y = \cos x$ (c) $y = \frac{1}{2}x \sin x$ (d) $y = -\frac{1}{2}x \cos x$

Exercise 3.0.6. Use Euler's method with step size 0.5 to compute the approximate y-values y_1, y_2, y_3, and y_4 of the solution of the initial-value problem

$$y' = y - 2x, \qquad y(1) = 0.$$

Exercise 3.0.7. Use Euler's method with step size 0.2 to estimate $y(1)$, where $y(x)$ is the solution of the initial-value problem

$$y' = xy - x^2, \qquad y(0) = 1.$$

Exercise 3.0.8. Use Euler's method with step size 0.1 to estimate $y(0.5)$, where $y(x)$ is the solution of the initial-value problem

$$y' = y + xy, \qquad y(0) = 1.$$

Exercise 3.0.9. (a) Use Euler's method with step size 0.2 to estimate $y(0.4)$, where $y(x)$ is the solution of the initial-value problem

$$y' = x + y^2, \qquad y(0) = 0.$$

(b) Repeat part (a) with step size 0.1.

3.1 Separable Equations

Exercise 3.1.1. Solve the differential equation.

(a) $\dfrac{dy}{dx} = xy^2$

(b) $\dfrac{dy}{dx} = xe^{-y}$

(c) $xy^2y' = x + 1$

(d) $\left(y^2 + xy^2\right) y' = 1$

(e) $(y + \sin y)y' = x + x^3$

(f) $\dfrac{dv}{ds} = \dfrac{s+1}{sv+s}$

(g) $\dfrac{dy}{dt} = \dfrac{t}{ye^{t+2}}$

(h) $\dfrac{dy}{d\theta} = \dfrac{e^y \sin^2\theta}{y \sec\theta}$

(i) $\dfrac{dp}{dt} = t^2p - p + t^2 - 1$

(j) $\dfrac{dz}{dt} + e^{t+z} = 0$

Exercise 3.1.2. Find the solution of the differential equation that satisfies the given initial condition.

(a) $\dfrac{dy}{dx} = \dfrac{x}{y}, \qquad y(0) = -3.$

(b) $\dfrac{dy}{dx} = \dfrac{\ln x}{xy}, \qquad y(1) = 2.$

(c) $\dfrac{du}{dt} = \dfrac{2t + \sec^2 t}{2u}, \qquad u(0) = -5.$

(d) $y' = \dfrac{xy \sin x}{y+1}, \qquad y(0) = 1.$

(e) $x \ln x = y\left(1 + \sqrt{3 + y^2}\right) y', \qquad y(1) = 1.$

(f) $\dfrac{dP}{dt} = \sqrt{Pt}, \qquad P(1) = 2.$

(g) $y' \tan x = a + y, \qquad y\left(\dfrac{\pi}{3}\right) = a, \quad 0 < x < \dfrac{\pi}{2}.$

(h) $\dfrac{dL}{dt} = kL^2 \ln t, \qquad L(1) = -1.$

Exercise 3.1.3. Find an equation of the curve that passes through the point $(0, 1)$ and whose slope at (x, y) is xy.

Exercise 3.1.4. Find the function f such that

$$f'(x) = f(x)\big(1 - f(x)\big), \qquad f(0) = \tfrac{1}{2}.$$

Solve the differential equation

$$y' = x + y$$

by making the change of variable

$$u = x + y.$$

Exercise 3.1.5. Solve the differential equation

$$xy' = y + xe^{y/x}$$

by making the change of variable

$$v = \frac{y}{x}.$$

Exercise 3.1.6. A sphere with radius 1 m has temperature 15°C. It lies inside a concentric sphere with radius 2 m and temperature 25°C. The temperature $T(r)$ at a distance r from the common center of the spheres satisfies the differential equation

$$\frac{d^2T}{dr^2} + \frac{2}{r}\frac{dT}{dr} = 0.$$

If we let $S = \dfrac{dT}{dr}$, then S satisfies a first-order differential equation. Solve it to find an expression for the temperature $T(r)$ between the spheres.

Exercise 3.1.7. A glucose solution is administered intravenously into the bloodstream at a constant rate r. As the glucose is added, it is converted into other substances and removed from the bloodstream at a rate that is proportional to the concentration at that time. Thus a model for the concentration $C = C(t)$ of the glucose solution in the bloodstream is

$$\frac{dC}{dt} = r - kC,$$

where k is a positive constant.

(a) Suppose that the concentration at time $t = 0$ is C_0. Determine the concentration at any time t by solving the differential equation.

(b) Assuming that $C_0 < \dfrac{r}{k}$, find $\lim\limits_{t\to\infty} C(t)$ and interpret the result.

3.2 Linear Equations

Exercise 3.2.1. Determine whether the differential equation is linear.

(a) $x - y' = xy$

(b) $y' + xy^2 = \sqrt{x}$

(c) $y' = \dfrac{1}{x} + \dfrac{1}{y}$

(d) $y \sin x = x^2 y' - x$

Exercise 3.2.2. Solve the differential equation.

(a) $y' + y = 1$

(b) $y' - y = e^x$

(c) $y' = x - y$

(d) $4x^3y + x^4y' = \sin^3 x$

(e) $xy' + y = \sqrt{x}$

(f) $y' + y = \sin(e^x)$

(g) $\sin x \dfrac{dy}{dx} + (\cos x)\, y = \sin(x^2)$

(h) $x\dfrac{dy}{dx} - 4y = x^4e^x$

(i) $(1+t)\dfrac{du}{dt} + u = 1 + t, \quad t > 0$

(j) $t \ln t \dfrac{dr}{dt} + r = te^t$

Exercise 3.2.3. Solve the initial-value problem

(a) $x^2y' + 2xy = \ln x, \qquad y(1) = 2.$

(b) $t^3\dfrac{dy}{dt} + 3t^2y = \cos t, \qquad y(\pi) = 0.$

(c) $t\dfrac{du}{dt} = t^2 + 3u, \qquad t > 0, \quad u(2) = 4.$

(d) $2xy' + y = 6x, \qquad x > 0, \quad y(4) = 20.$

(e) $xy' = y + x^2 \sin x, \qquad y(\pi) = 0.$

(f) $(x^2+1)\dfrac{dy}{dx} + 3x(y-1) = 0, \qquad y(0) = 2.$

3.3 Second-order Linear Differential Equations

Exercise 3.3.1. Solve the differential equation.

(a) $y'' - y' - 6y = 0$

(b) $y'' + 4y' + 4y = 0$

(c) $y'' + 16y = 0$

(d) $y'' - 8y' + 12y = 0$

(e) $9y'' - 12y' + 4y = 0$

(f) $25y'' + 9y = 0$

(g) $y' = 2y''$

(h) $y'' - 4y' + y = 0$

(i) $y'' - 4y' + 13y = 0$

(j) $y'' + 3y' = 0$

(k) $2\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} - y = 0$

(l) $8\dfrac{d^2y}{dt^2} + 12\dfrac{dy}{dt} + 5y = 0$

(m) $100\dfrac{d^2P}{dt^2} + 200\dfrac{dP}{dt} + 101P = 0$

Exercise 3.3.2. Solve the initial-value problem.

(a) $y'' - 6y' + 8y = 0, \qquad y(0) = 2, \quad y'(0) = 2$

(b) $y'' + 4y = 0, \qquad y(\pi) = 5, \quad y'(\pi) = -4$

(c) $9y'' + 12y' + 4y = 0, \qquad y(0) = 1, \quad y'(0) = 0$

(d) $2y'' + y' - y = 0, \qquad y(0) = 3, \quad y'(0) = 3$

(e) $y'' - 6y' + 10y = 0, \qquad y(0) = 2, \quad y'(0) = 3$

(f) $4y'' - 20y' + 25y = 0, \qquad y(0) = 2, \quad y'(0) = -3$

(g) $y'' - y' - 12y = 0, \qquad y(1) = 0, \quad y'(1) = 1$

(h) $4y'' + 4y' + 3y = 0, \qquad y(0) = 0, \quad y'(0) = 1$

Exercise 3.3.3. Solve the boundary-value problem, if possible

(a) $y'' + 4y = 0, \qquad y(0) = 5, \quad y\left(\frac{\pi}{4}\right) = 3$

(b) $y'' = 4y, \qquad y(0) = 1, \quad y(1) = 0$

(c) $y'' + 4y' + 4y = 0, \qquad y(0) = 2, \quad y(1) = 0$

(d) $y'' - 8y' + 17y = 0, \qquad y(0) = 3, \quad y(\pi) = 2$

(e) $y'' = y', \qquad y(0) = 1, \quad y(1) = 2$

(f) $4y'' - 4y' + y = 0, \qquad y(0) = 4, \quad y(2) = 0$

(g) $y'' + 4y' + 20y = 0, \qquad y(0) = 1, \quad y(\pi) = 2$

(h) $y'' + 4y' + 20y = 0, \qquad y(0) = 1, \quad y(\pi) = e^{-2\pi}$

Exercise 3.3.4. Solve the differential equation or initial-value problem using the method of undetermined coefficients.

(a) $y'' - 2y' - 3y = \cos 2x$

(b) $y'' - y = x^3 - x$

(c) $y'' + 9y = e^{-2x}$

(d) $y'' + 2y' + 5y = 1 + e^x$

(e) $y'' - 4y' + 5y = e^{-x}$

(f) $y'' - 4y' + 4y = x - \sin x$

(g) $y'' + y = e^x + x^3, \qquad y(0) = 2, \quad y'(0) = 0$

(h) $y'' - 4y = e^x \cos x, \qquad y(0) = 1, \quad y'(0) = 2$

(i) $y'' - y' = xe^x, \qquad y(0) = 2, \quad y'(0) = 1$

(j) $y'' + y' - 2y = x + \sin 2x, \qquad y(0) = 1, \quad y'(0) = 0$

Chapter 4

Vector and geometry of space

4.1 Vectors

Exercise 4.1.1. Find $\boldsymbol{a}+\boldsymbol{b}$, $2\boldsymbol{a}+3\boldsymbol{b}$, $|\boldsymbol{a}|$ and $|\boldsymbol{a}-\boldsymbol{b}|$.

(a) $\boldsymbol{a}=(5,-12)$, $\boldsymbol{b}=(-3,-6)$.

(b) $\boldsymbol{a}=4\boldsymbol{i}+\boldsymbol{j}$, $\boldsymbol{b}=\boldsymbol{i}-2\boldsymbol{j}$,

(c) $\boldsymbol{a}=\boldsymbol{i}+2\boldsymbol{j}-3\boldsymbol{k}$, $\boldsymbol{b}=-2\boldsymbol{i}-\boldsymbol{j}+5\boldsymbol{k}$,

(d) $\boldsymbol{a}=2\boldsymbol{i}-4\boldsymbol{j}+4\boldsymbol{k}$, $\boldsymbol{b}=2\boldsymbol{j}-\boldsymbol{k}$.

Exercise 4.1.2. Find a unit vector that has the same direction as the given vector.

(a) $\boldsymbol{a}=-3\boldsymbol{i}+7\boldsymbol{j}$,

(b) $\boldsymbol{b}=(-4,2,4)$,

(c) $\boldsymbol{a}=8\boldsymbol{i}-\boldsymbol{j}+4\boldsymbol{k}$,

Exercise 4.1.3. Find a vector that has the same direction as $(-2,4,2)$ but has length 6.

Exercise 4.1.4. A boatman wants to cross a canal that is 3 km wide and wants to land at a point 2 km upstream from his starting point. The current in the canal flows at 3.5 km/h and the speed of his boat is 13 km/h.

(a) In what direction should he steer?

(b) How long will the trip take?

Exercise 4.1.5. Find $\boldsymbol{a}\cdot\boldsymbol{b}$.

(a) $\boldsymbol{a}=(-2,3)$, $\quad\boldsymbol{b}=(0.7,1.2)$

(b) $\boldsymbol{a}=(6,-2,3)$, $\quad\boldsymbol{b}=(2,5,-1)$

(c) $\boldsymbol{a}=2\boldsymbol{i}+\boldsymbol{j}$, $\quad\boldsymbol{b}=\boldsymbol{i}-\boldsymbol{j}+\boldsymbol{k}$

(d) $|\boldsymbol{a}|=6$, $\quad|\boldsymbol{b}|=5$, the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$ is $2\pi/3$.

(e) $|\boldsymbol{a}|=3$, $\quad|\boldsymbol{b}|=\sqrt{6}$, the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$ is 45°.

Exercise 4.1.6. Find the angle between the vectors.

(a) $\boldsymbol{a} = (4, 3), \quad \boldsymbol{b} = (2, -1)$

(b) $\boldsymbol{a} = (3, -1, 5), \quad \boldsymbol{b} = (-2, 4, 3)$

(c) $\boldsymbol{a} = 4\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}, \quad \boldsymbol{b} = 2\boldsymbol{i} - \boldsymbol{k}$

Exercise 4.1.7. Determine whether the given vectors are orthogonal, parallel, or neither.

(a) $\boldsymbol{a} = (-5, 3, 7), \quad \boldsymbol{b} = (6, -8, 2)$

(b) $\boldsymbol{a} = (4, 6), \quad \boldsymbol{b} = (-3, 2)$

(c) $\boldsymbol{a} = -\boldsymbol{i} + 2\boldsymbol{j} + 5\boldsymbol{k}, \quad \boldsymbol{b} = 3\boldsymbol{i} + 4\boldsymbol{j} - \boldsymbol{k}$

Exercise 4.1.8. Find the scalar and vector projections of onto .

(a) $\boldsymbol{a} = (-5, 12), \quad \boldsymbol{b} = (4, 6)$

(b) $\boldsymbol{a} = (3, 6, -2), \quad \boldsymbol{b} = (1, 2, 3)$

(c) $\boldsymbol{a} = 2\boldsymbol{i} - \boldsymbol{j} + 4\boldsymbol{k}, \quad \boldsymbol{b} = \boldsymbol{j} + \frac{1}{2}\boldsymbol{k}$

Exercise 4.1.9. Find the cross product $\boldsymbol{a} \times \boldsymbol{b}$ and verify that it is orthogonal to both $\boldsymbol{a}$ and $\boldsymbol{b}$.

(a) $\boldsymbol{a} = (6, 0, -2), \quad \boldsymbol{b} = (0, 8, 0)$

(b) $\boldsymbol{a} = (1, 1, -1), \quad \boldsymbol{b} = (2, 4, 6)$

(c) $\boldsymbol{a} = \boldsymbol{i} + 3\boldsymbol{j} - 2\boldsymbol{k}, \quad \boldsymbol{b} = -\boldsymbol{i} + 5\boldsymbol{k}$

(d) $\boldsymbol{a} = t\boldsymbol{i} + \cos t\,\boldsymbol{j} + \sin t\,\boldsymbol{k}, \quad \boldsymbol{b} = \boldsymbol{i} - \sin t\,\boldsymbol{j} + \cos t\,\boldsymbol{k}$

4.2 Equations of Lines and Planes

Exercise 4.2.1. Find a vector equation and parametric equations for the line.

(a) The line through the point $(6, -5, 2)$ and parallel to the vector $(1, 3, -\frac{2}{3})$.

(b) The line through the point $(2, 2.4, 3.5)$ and parallel to the vector $3\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

(c) The line through the point $(0, 14, -10)$ and parallel to the line $x = -1 + 2t$, $y = 6 - 3t$, $z = 3 + 9t$.

(d) The line through the point $(1, 0, 6)$ and perpendicular to the plane $x + 3y + z = 5$.

Exercise 4.2.2. Find parametric equations and symmetric equations for the line.

(a) The line through the points $(0, \frac{1}{2}, 1)$ and $(2, 1, -3)$.

(b) The line through the point $(2, 1, 0)$ and perpendicular to both $\boldsymbol{i} + \boldsymbol{j}$ and $\boldsymbol{j} + \boldsymbol{k}$.

(c) The line through the point $(1, -1, 1)$ and parallel to the line $x + 2 = \frac{1}{2}y = z - 3$.

(d) The line of intersection of the planes $x + 2y + 3z = 1$ and $x - y + z = 1$.

Exercise 4.2.3. Determine whether the lines L_1 and L_2 are parallel, skew, or intersecting. If they intersect, find the point of intersection.

(a)

$$\begin{aligned} L_1: &\quad x = 3 + 2t, \quad y = 4 - t, \quad z = 1 + 3t \\ L_2: &\quad x = 1 + 4s, \quad y = 3 - 2s, \quad z = 4 + 5s \end{aligned}$$

(b)

$$\begin{aligned} L_1: &\quad x = 5 - 12t, \quad y = 3 + 9t, \quad z = 1 - 3t \\ L_2: &\quad x = 3 + 8s, \quad y = -6s, \quad z = 7 + 2s \end{aligned}$$

(c)

$$\begin{aligned} L_1: &\quad \frac{x-2}{1} = \frac{y-3}{-2} = \frac{z-1}{-3} \\ L_2: &\quad \frac{x-3}{1} = \frac{y+4}{3} = \frac{z-2}{-7} \end{aligned}$$

(d)

$$\begin{aligned} L_1: &\quad \frac{x}{1} = \frac{y-1}{-1} = \frac{z-2}{3} \\ L_2: &\quad \frac{x-2}{2} = \frac{y-3}{-2} = \frac{z}{7} \end{aligned}$$

Exercise 4.2.4. Find an equation of the plane.

(a) The plane through the origin and perpendicular to the vector $\langle 1, -2, 5 \rangle$.

(b) The plane through the point $(5, 3, 5)$ and with normal vector $2\mathbf{i} + \mathbf{j} - \mathbf{k}$.

(c) The plane through the point $\left(-1, \frac{1}{2}, 3\right)$ and with normal vector $\mathbf{i} + 4\mathbf{j} + \mathbf{k}$.

(d) The plane through the point $(2, 0, 1)$ and perpendicular to the line

$$x = 3t, \quad y = 2 - t, \quad z = 3 + 4t$$

(e) The plane through the point $(1, -1, -1)$ and parallel to the plane

$$5x - y - z = 6$$

(f) The plane through the point $(2, 4, 6)$ and parallel to the plane

$$z = x + y$$

(g) The plane through the point $\left(1, \frac{1}{2}, \frac{1}{3}\right)$ and parallel to the plane

$$x + y + z = 0$$

(h) The plane that contains the line

$$x = 1 + t, \quad y = 2 - t, \quad z = 4 - 3t$$

and is parallel to the plane

$$5x + 2y + z = 1$$

(i) The plane through the points $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$.

(j) The plane through the origin and the points $(2, -4, 6)$ and $(5, 1, 3)$.

(k) The plane through the points $(3, -1, 2), (8, 2, 4)$, and $(-1, -2, -3)$.

(l) The plane that passes through the point $(1, 2, 3)$ and contains the line

$$x = 3t, \quad y = 1 + t, \quad z = 2 - t$$

(m) The plane that passes through the point $(6, 0, -2)$ and contains the line

$$x = 4 - 2t, \quad y = 3 + 5t, \quad z = 7 + 4t$$

(n) The plane that passes through the point $(1, -1, 1)$ and contains the line with symmetric equations

$$x = 2y = 3z$$

(o) The plane that passes through the point $(-1, 2, 1)$ and contains the line of intersection of the planes

$$x + y - z = 2$$

and

$$2x - y + 3z = 1$$

(p) The plane that passes through the points $(0, -2, 5)$ and $(-1, 3, 1)$ and is perpendicular to the plane

$$2z = 5x + 4y$$

(q) The plane that passes through the point $(1, 5, 1)$ and is perpendicular to the planes

$$2x + y - 2z = 2 \quad \text{and} \quad x + 3z = 4$$

(r) The plane that passes through the line of intersection of the planes

$$x - z = 1 \quad \text{and} \quad y + 2z = 3$$

and is perpendicular to the plane

$$x + y - 2z = 1$$

4.3 Three-Dimensional Coordinate Systems

Exercise 4.3.1. Find the lengths of the sides of the triangle PQR. Is it a right triangle? Is it an isosceles triangle?

(a) $P(3, -2, -3)$, $Q(7, 0, 1)$, $R(1, 2, 1)$,

(b) $P(2, -1, 0)$, $Q(4, 1, 1)$, $R(4, -5, 4)$.

Exercise 4.3.2. Determine whether the points lie on straight line.

(a) $A(2,4,2)$, $B(3,7,-2)$, $C(1,3,3)$,

(b) $D(0,-5,5)$, $E(1,-2,4)$, $F(3,4,2)$.

Exercise 4.3.3. Find an equation of the sphere with center $(-3,2,5)$ and radius 4. What is the intersection of this sphere with the yz-plane?

Exercise 4.3.4. Find an equation of the sphere that passes through the point $(4,3,-1)$ and has center $(3,8,1)$.

Exercise 4.3.5. Show that the equation represents a sphere, and find its center and radius

(a) $x^2+y^2+z^2-2x-4y+8z=15$

(b) $x^2+y^2+z^2+8x-6y+2z+17=0$

(c) $2x^2+2y^2+2z^2=8x-24z+1$

(d) $3x^2+3y^2+3z^2=10+6y+12z$

Chapter 5

Partial Derivatives

5.1 Limits

Exercise 5.1.1. Find the limit, if it exists, or show that the limit does not exist

(a) $\lim\limits_{(x,y)\to(1,2)} (5x^3 - x^2y^2)$

(b) $\lim\limits_{(x,y)\to(1,-1)} e^{-xy}\cos(x+y)$

(c) $\lim\limits_{(x,y)\to(2,1)} \dfrac{4-xy}{x^2+3y^2}$

(d) $\lim\limits_{(x,y)\to(0,0)} \dfrac{y^4}{x^4+3y^4}$

(e) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2+\sin^2 y}{2x^2+y^2}$

(f) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2+y^2}}$

(g) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^4-y^4}{x^2+y^2}$

(h) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2ye^y}{x^4+4y^2}$

(i) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2\sin^2 y}{x^2+2y^2}$

(j) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^4}{x^2+y^8}$

Exercise 5.1.2. Determine if the limit exists

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

Exercise 5.1.3. Determine if the limit exists

$$\lim_{(x,y)\to(0,0)} \frac{xy^4}{x^2+y^4}$$

Exercise 5.1.4. Determine if the limit exists

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}$$

Exercise 5.1.5. Use polar coordinates to find the following limits

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^3+y^3}{x^2+y^2}$

(b) $\lim\limits_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2)$

(c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^{-x^2-y^2}-1}{x^2+y^2}$

5.2 Continuity

Exercise 5.2.1. Determine the set of points at which the function is continuous.

(a) $F(x,y) = \dfrac{\sin(xy)}{e^x - y^2}$

(b) $F(x,y) = \dfrac{x-y}{1+x^2+y^2}$

(c) $F(x,y) = e^{x^2 y} + \sqrt{x+y^2}$

(d) $F(x,y) = \ln(x^2+y^2-4)$

(e) $F(x,y) = \sqrt{x+y+z}$

Exercise 5.2.2. Determine if f is continuous at $(0,0)$

$$f(x,y) = \begin{cases} \dfrac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Exercise 5.2.3. Determine if g is continuous at $(0,0)$

$$g(x,y) = \begin{cases} \dfrac{3x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Exercise 5.2.4. Determine if g is continuous at $(0,0)$

$$g(x,y) = \begin{cases} \dfrac{x^2y^3}{2x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 1 & \text{if } (x,y) = (0,0) \end{cases}$$

5.3 Partial derivatives

Exercise 5.3.1. Find the first partial derivatives of the function.

(a) $f(x,y) = y^5 - 3xy$

(b) $f(x,y) = x^4y^3 + 8x^2y$

(c) $f(x,t) = e^{-t}\cos \pi x$

(d) $f(x,t) = \sqrt{x}\ln t$

(e) $f(x,y) = x\ln(x^2+y^2)$

(f) $u = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}$

(g) $u = \sin(x_1 + 2x_2 + \ldots + nx_n)$

(h) $u = x^{y/z}$

Exercise 5.3.2. Find the indicated partial derivatives

(a) $f(x,y) = \ln\left(x + \sqrt{x^2+y^2}\right)$; $\quad f_x(3,4)$

(b) $f(x,y) = \arctan(\mathrm{y/x})$; $\quad \mathrm{f_x}(2,3)$

(c) $f(x,y,z) = \dfrac{y}{x+y+z}$; $\quad f_y(2,1,-1)$

(d) $f(x,y,z) = \sqrt{\sin^2 x + \sin^2 y + \sin^2 z}$; $\quad f_z(0,0,\pi/4)$

Exercise 5.3.3. Find all the second partial derivatives

(a) $f(x,y) = x^3y^5 + 2x^4y$

(b) $f(x,y) = \sin^2(mx + ny)$

(c) $w = \sqrt{u^2 + v^2}$

(d) $v = e^{xe^y}$

Exercise 5.3.4. Use definition to find the partial derivatives $f_x(x,y)$ and $f_y(x,y)$

(a) $f(x,y) = xy^2 - x^3y$

(b) $f(x,y) = \dfrac{x}{x+y^2}$

Exercise 5.3.5. Determine whether each of the following functions is a solution of the Laplace's equation $u_{xx} + u_{yy} = 0$.

(a) $u = x^2 + y^2$

(b) $u = x^2 - y^2$

(c) $u = x^3 + 3xy^2$

(d) $u = \ln\sqrt{x^2 + y^2}$

Exercise 5.3.6. Verify that the function $u = 1/\sqrt{x^2 + y^2 + z^2}$ is a solution of the three-dimensional Laplace equation $u_{xx} + u_{yy} + u_{zz} = 0$.

Exercise 5.3.7. Let

$$f(x,y) = \begin{cases} \dfrac{x^3y - xy^3}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Prove that f is continuous in $\mathbb{R}^2$.

(b) Find $f_x(x,y)$ and $f_y(x,y)$ when $(x,y) \neq (0,0)$.

(c) Find $f_x(0,0)$ and $f_y(0,0)$.

(d) Show that $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = 1$.

Exercise 5.3.8. Find the equation of the tangent plane to the given surface at the specified point.

(a) $z = 4x^2 - y^2 + 2y, \quad (-1,2,4)$

(b) $z = 3(x-1)^2 + 2(y+3)^2 + 7, \quad (2,-2,12)$

(c) $z = y\ln x, \quad (1,4,0)$

Exercise 5.3.9. Explain why the function is differentiable at the given point. Then find the linearization $L(x,y)$ of the function at that point.

(a) $f(x,y) = x\sqrt{y}, \quad (1,4)$

(b) $f(x,y) = x^3y^4, \quad (1,1)$

(c) $f(x,y) = \frac{x}{x+y} \quad (2,1)$

Exercise 5.3.10. Find the linear approximation of the function

$$f(x,y) = \sqrt{20 - x^2 - 7y^2}$$

at $(2,1)$ and use it to approximate $f(1.95, 1.08)$.

Exercise 5.3.11. Let

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Show that that $f_x(0,0)$ and $f_y(0,0)$ both exist but f is not differentiable at $(0,0)$.

(b) Prove that f_x and f_y are not continuous at $(0,0)$.

Exercise 5.3.12. Use the chain rule to find dz/dt or du/dt

(a) $z = x^2 + y^2 + xy, \quad x = \sin t, \quad y = e^t$

(b) $z = \cos(x + 4y), \quad x = 5t^4, \quad y = 1/t$

(c) $z = \sqrt{1 + x^2 + y^2}, \quad x = \ln t, \quad y = \cos t$

(d) $u = \ln \sqrt{x^2 + y^2 + z^2}, \quad x = \sin t, \quad y = \cos t, \quad z = \tan t.$

Exercise 5.3.13. Use the chain rule to find $\partial z/\partial s$ and $\partial z/\partial t$.

(a) $z = x^2y^2, \quad x = s\cos t, \quad y = s\sin t$

(b) $z = \sin\theta\, \cos\varphi, \quad \theta = st^2, \quad \varphi = s^2t$

(c) $z = e^r\, \cos\theta, \quad r = st, \quad \theta = \sqrt{s^2 + t^2}.$

Exercise 5.3.14. If $z = f(x,y)$, where f is differentiable, and

$$\begin{aligned} x &= g(t) & y &= h(t) \\ g(3) &= 2 & h(3) &= 7 \\ g'(3) &= 5 & h'(3) &= -4 \\ f_x(2,7) &= 6 & f_y(2,7) &= -8, \end{aligned}$$

find dz/dt when $t = 3$.

5.4 Local and absolute Extremums

Exercise 5.4.1. Find the local maximum and minimum values and saddle point(s) of the functions

(a) $f(x,y) = 9 - 2x + 4y - x^2 - 4y^2$

(b) $f(x,y) = x^3y + 12x^2 - 8y$

(c) $f(x,y) = x^4 + y^4 - 4xy + 2$

(d) $f(x,y) = (1 + xy)(x + y)$

Exercise 5.4.2. Find the absolute maximum and minimum values of f on the set Ω.

(a) $f(x,y) = 1 + 4x - 5y$ and Ω is the closed triangular region with vertices $(0,0)$, $(2,0)$ and $(0,3)$.

(b) $f(x,y) = 3+xy-x-2y$ and Ω is the closed triangular region with vertices $(1,0)$, $(5,0)$ and $(1,4)$.

(c) $f(x,y) = x^2+y^2+x^2y+4$ and $\Omega = \{(x,y) : |x| \leq 1,\ |y| \leq 1\}$.

(d) $f(x,y) = 4x+6y-x^2-y^2$ and $\Omega = [0,4] \times [0,5]$

(e) $f(x,y) = x^4+y^4-x^2-y^2$ and $\Omega = [0,3] \times [0,2]$

(f) $f(x,y) = xy^2$ and $\Omega = \{(x,y) : x \geq 0, y \geq 0, x^2+y^2 \leq 3\}$

(g) $f(x,y) = 2x^3+y^4$ and $\Omega = \{(x,y) : x^2+y^2 \leq 1\}$

Exercise 5.4.3. Find the point on the plane $x-y+z=4$ that is closest to the point $(1,2,3)$.

Exercise 5.4.4. Find the point on the surface $z^2 = x^2+y^2$ that is closest to the point $(1,2,3)$.

Exercise 5.4.5. Find three posible numbers whose sum is 100 and whose product is a maximum.

Exercise 5.4.6. Find the dimensions of the box with volume 1000 cm^3 that has minimal surface area.

Exercise 5.4.7. The base of an aquarium with given volume V is made of slate and the sides are made of glass. If slate costs five times as much (per unit area) as glass, find the dimensions of the aquarium that minimize the cost of the materials

Exercise 5.4.8. Find the maximum and minimum values of the functions subject to the given constraint

(a) $f(x,y) = x^2+y^2; \quad xy = 1$

(b) $f(x,y) = 4x+6y; \quad x^2+y^2 = 13$

(c) $f(x,y,z) = x^2y^2z^2; \quad x^2+y^2+z^2 = 1$

(d) $f(x,y,z) = xyz; \quad x^2+2y^2+3z^2 = 6$

Chapter 6

Multiple Integrals

6.1 Double Integrals

Exercise 6.1.1. Estimate the volume of the solid that lies below the surface $z = x + 2y^2$ and above the rectangle $R = [0, 2] \times [0, 4]$.

Exercise 6.1.2. Evaluate the double integrals

(a) $\displaystyle\iint_R 3\,dA, \quad R = \{(x, y) : -2 \le x \le 2,\ 1 \le y \le 6\}$

(b) $\displaystyle\iint_R (5 - x)\,dA, \quad R = \{(x, y) : 0 \le x \le 5,\ 0 \le y \le 3\}$

Exercise 6.1.3. Prove that

$$0 \le \iint_R \sin \pi x \, \cos \pi y \le \frac{1}{32}$$

where $R = \left[0, \frac{1}{4}\right] \times \left[\frac{1}{4}, \frac{1}{2}\right]$.

Exercise 6.1.4. Calculate the following integrals

(a) $\displaystyle\int_1^3 \int_0^1 (1 + 4xy)\,dx\,dy$

(b) $\displaystyle\int_0^1 \int_1^2 (4x^3 - 9x^2y^2)\,dy\,dx$

(c) $\displaystyle\int_0^2 \int_0^{\pi/2} x \sin y\,dy\,dx$

(d) $\displaystyle\int_{\pi/6}^{\pi/2} \int_{-1}^5 \cos y\,dy\,dx$

(e) $\displaystyle\int_0^1 \int_0^1 xy\sqrt{x^2 + y^2}\,dy\,dx$

Exercise 6.1.5. Calculate the following double integrals

(a) $\displaystyle\iint_R (6x^2y^3 - 5y^4)\,dA, \quad R = [0, 3] \times [0, 1]$

(b) $\displaystyle\iint_R \cos(x + 2y)\,dA, \quad R = [0, \pi] \times [0, \pi/2]$

(c) $\displaystyle\iint_R \frac{xy^2}{x^2 + 1}\,dA, \quad R = [0, 1] \times [-3, 3]$

(d) $\displaystyle\iint_R \frac{1+x^2}{1+y^2}\,dA, \quad R=[0,1]\times[0,1]$

(e) $\displaystyle\iint_R xye^{x^2y}\,dA, \quad R=[0,1]\times[0,2].$

Exercise 6.1.6. Find the volume of the solid that lies under the plane $3x+2y+z=12$ and above the rectangle $R=[0,1]\times[-2,3]$.

Exercise 6.1.7. Find the volume of the solid that lies under the hyperbolic paraboloid $z=4+x^2-y^2$ and above the square $R=[-1,1]\times[0,2]$.

Exercise 6.1.8. Find the volume of the solid enclosed by the surface $z=1+e^x\sin y$ and the planes $x=\pm1$, $y=0$, $y=\pi$

Exercise 6.1.9. Evaluate the double integrals

(a) $\displaystyle\iint_D y^2\,dA, \quad D=\{(x,y):-1\le y\le 1,\ -y-2\le x\le y\}$

(b) $\displaystyle\iint_D \frac{y}{x^5+1}\,dA, \quad D=\{(x,y):0\le x\le 1,\ 0\le y\le x^2\}$

(c) $\displaystyle\iint_D x\,dA, \quad D=\{(x,y):0\le x\le \pi,\ 0\le y\le \sin x\}$

(d) $\displaystyle\iint_D x\cos y\,dA,$ D is bounded by $y=0$, $y=x^2$ and $x=1$

(e) $\displaystyle\iint_D (x+y)\,dA,$ D is bounded by $y=\sqrt{x}$, $y=x^2$

(f) $\displaystyle\iint_D y^3\,dA,$ D is the triangular region with vertices $(0,2)$, $(1,1)$ and $(3,2)$

(g) $\displaystyle\iint_D xy^2\,dA,$ D is enclosed by $x=0$ and $x=\sqrt{1-y^2}$

Exercise 6.1.10. Find the volume of the given solid

(a) Under the plane $x+2y-z=0$ and above the region bounded by $y=x$ and $y=x^4$.

(b) Under the surface $z=2x+y^2$ and above the region bounded by $x=y^2$ and $x=y^3$.

(c) Bounded by the plane $z=x$, $y=x$, $x+y=2$ and $z=0$.

(d) in the half space $x\ge 0$, bounded by the cyclinders $z=x^2$, $y=x^2$ and the planes $z=0$, $y=4$.

6.2 Polar coordinates - Double integrals

Exercise 6.2.1. Evaluate the integrals by changing into polar coordinates.

(a) $\displaystyle\iint_D xy\,dA$, where D is the disk with center the origin and radius 3

(b) $\iint_R (x+y)\,dA$, where R is the region that lies left of the y-axis between the circles $x^2+y^2=1$ and $x^2+y^2=4$.

(c) $\iint_R \sqrt{4-x^2-y^2}\,dA$, where $R=\{(x,y): x^2+y^2\le 4, x\ge 0\}$

(d) $\iint_R (2x-y)\,dA$, where R is the region in the first quadrant enclosed by the circle $x^2+y^2=4$ and the lines $x=0$ and $y=x$.

(e) $\iint_D e^{-x^2-y^2}\,dA$, where D is the region bounded by the semicircle $x=\sqrt{4-y^2}$ and the y-axis.

6.3 Triple Integrals

Exercise 6.3.1. Evaluate the integral $\iiint_E (xy+z^2)\,\mathrm{dV}$, where

$$E=\{(x,y,z): 0\le x\le 2, 0\le y\le 1, 0\le z\le 3\}.$$

Exercise 6.3.2. Evaluate the iterated integrals.

(a) $\int_0^2\int_0^{z^2}\int_0^{y-z}(2x-y)\,dx\,dy\,dz$

(b) $\int_1^2\int_0^{2z}\int_0^{\ln x} xe^{-y}\,dy\,dx\,dz$

(c) $\int_0^{\pi/2}\int_0^{y}\int_0^{x}\cos(x+y+z)\,dz\,dx\,dy$

Exercise 6.3.3. Evaluate triple integrals.

(a) $\iiint_E y\,dV$, where $E=\{(x,y,z): 0\le x\le 2, 0\le y\le x, x-y\le z\le x+y\}$.

(b) $\iiint_E e^{z/y}\,dV$, where $E=\{(x,y,z): 0\le y\le 1, y\le x\le 1, 0\le z\le xy\}$.

(c) $\iiint_E \frac{z}{x^2+z^2}\,dV$, where $E=\{(x,y,z): 1\le y\le 4, y\le z\le 4, 0\le x\le z\}$.

Exercise 6.3.4. Evaluate $\iiint_E \sin y\,dV$, where E lies below the plane $z=x$ and above the triangular region with vertices $(0,0,0)$, $(\pi,0,0)$, and $(0,\pi,0)$.

Exercise 6.3.5. Evaluate $\iiint_E 6xy\,dV$, where E lies under the plane $z=1+x+y$ and above the region in the xy-plane bounded by the curves $y=\sqrt{x}$, $y=0$, and $x=1$.

Exercise 6.3.6. Evaluate $\iiint_E xy\,dV$, where E is bounded by the parabolic cylinders $y=x^2$ and $x=y^2$ and the planes $z=0$ and $z=x+y$.

Exercise 6.3.7. Evaluate $\iiint_T x^2\,dV$, where T is the solid tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Exercise 6.3.8. Evaluate $\iiint_E x\,dV$, where E is bounded by the paraboloid $x=4y^2+4z^2$ and the plane $x=4$.

6.4 Cylindrical coordinates - Triple integrals

Important formulas:

$$\iiint_E f(x,y,z)\,dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z)\, r\,dz\,dr\,d\theta$$

$$E = \{(x,y,z) \mid (x,y) \in D,\ u_1(x,y) \le z \le u_2(x,y)\}$$

where D is given in polar coordinates by

$$D = \{(r,\theta) \mid \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$$

Exercise 6.4.1. Evaluate $\iiint_E \sqrt{x^2+y^2}\,dV$, where E is the region that lies inside the cylinder $x^2+y^2=16$ and between the planes $z=-5$ and $z=4$.

Exercise 6.4.2. Evaluate $\iiint_E z\,dV$, where E is enclosed by the paraboloid $z=x^2+y^2$ and the plane $z=4$.

Exercise 6.4.3. Evaluate $\iiint_E (x+y+z)\,dV$, where E is the solid in the first octant that lies under the paraboloid $z=4-x^2-y^2$.

Exercise 6.4.4. Evaluate $\iiint_E x\,dV$, where E is enclosed by the planes $z=0$ and $z=x+y+5$ and by the cylinders $x^2+y^2=4$ and $x^2+y^2=9$.

Exercise 6.4.5. Evaluate $\iiint_E x^2\,dV$, where E is the solid that lies within the cylinder $x^2+y^2=1$, above the plane $z=0$, and below the cone $z^2=4x^2+4y^2$.

Exercise 6.4.6. Find the volume of the solid that lies within both the cylinder $x^2+y^2=1$ and the sphere $x^2+y^2+z^2=4$.

Exercise 6.4.7. Find the volume of the solid that is enclosed by the cone $z=\sqrt{x^2+y^2}$ and the sphere $x^2+y^2+z^2=2$.

6.5 Spherical coordinates

Important formulas:

$$\iiint_E f(x,y,z)\,dV = \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\,\rho^2\sin\phi\,d\rho\,d\theta\,d\phi$$

where E is a spherical wedge given by

$$E = \{(\rho,\theta,\phi) \mid a \le \rho \le b,\ \alpha \le \theta \le \beta,\ c \le \phi \le d\}$$

Exercise 6.5.1. Evaluate $\iiint_E (x^2+y^2+z^2)^2\,dV$, where E is the ball with center the origin and radius 5.

Exercise 6.5.2. Evaluate $\iiint_H (9 - x^2 - y^2)\, dV$, where H is the solid hemisphere $x^2 + y^2 + z^2 \leq 9$, $z \geq 0$.

Exercise 6.5.3. Evaluate $\iiint_E (x^2 + y^2)^2\, dV$, where E lies between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

Exercise 6.5.4. Evaluate $\iiint_E y^2\, dV$, where E is the solid hemisphere $x^2 + y^2 + z^2 \leq 9$ and $y \geq 0$.

Exercise 6.5.5. Evaluate $\iiint\limits_E xe^{x^2+y^2+z^2}\, dV$, where E is the portion of the unit ball $x^2 + y^2 + z^2 \leq 1$ that lies in the first octant.

Exercise 6.5.6. Evaluate $\iiint\limits_E xyz\, dV$, where E lies between the spheres $\rho = 2$ and $\rho = 4$ and above the cone $\phi = \pi/3$.

Exercise 6.5.7. Find the volume of the part of the ball $\rho \leq a$ that lies between the cones $\phi = \pi/6$ and $\phi = \pi/3$.

Exercise 6.5.8. Find the average distance from a point in a ball of radius a to its center.

Exercise 6.5.9. (a) Find the volume of the solid that lies above the cone $\phi = \pi/3$ and below the sphere $\rho = 4\cos\phi$.

(b) Find the centroid of the solid in part (a).

Exercise 6.5.10. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy-plane, and below the cone $z = \sqrt{x^2 + y^2}$.

Chapter 7

Vector Calculus

7.1 Vector Fields

Exercise 7.1.1. Sketch the vector field $\mathbf{F}$ by drawing a diagram.

(a) $\mathbf{F}(x, y) = 0.3\,\mathbf{i} - 0.4\,\mathbf{j}$

(b) $\mathbf{F}(x, y) = \frac{1}{2}x\,\mathbf{i} + y\,\mathbf{j}$

(c) $\mathbf{F}(x, y) = -\frac{1}{2}\,\mathbf{i} + (y - x)\,\mathbf{j}$

(d) $\mathbf{F}(x, y) = y\,\mathbf{i} + (x + y)\,\mathbf{j}$

(e) $\mathbf{F}(x, y) = \dfrac{y\,\mathbf{i} + x\,\mathbf{j}}{\sqrt{x^2 + y^2}}$

(f) $\mathbf{F}(x, y) = \dfrac{y\,\mathbf{i} - x\,\mathbf{j}}{\sqrt{x^2 + y^2}}$

(g) $\mathbf{F}(x, y, z) = \mathbf{k}$

(h) $\mathbf{F}(x, y, z) = -y\,\mathbf{k}$

(i) $\mathbf{F}(x, y, z) = x\,\mathbf{k}$

(j) $\mathbf{F}(x, y, z) = \mathbf{j} - \mathbf{i}$

Exercise 7.1.2. Find the gradient vector field of f.

(a) $f(x, y) = xe^{xy}$

(b) $f(x, y) = \tan(3x - 4y)$

(c) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

(d) $f(x, y, z) = x \ln(y - 2z)$.

Exercise 7.1.3. Sketch the vector field by drawing a diagram

(a) $\boldsymbol{F}(x, y) = 0.3\boldsymbol{i} + 0.4\boldsymbol{j}$

(b) $\boldsymbol{F}(x, y) = \frac{1}{2}x\boldsymbol{i} + y\boldsymbol{j}$

(c) $\boldsymbol{F}(x, y, z) = \boldsymbol{k}$

(d) $\boldsymbol{F}(x, y, z) = x\boldsymbol{k}$

7.2 Line Integrals

Important formulas:

- A plane curve C is given by the parametric equations

$$x = x(t) \qquad y = y(t) \qquad a \le t \le b,$$

and f is a continuous. Then

$$\int_C f(x,y)\,ds = \int_a^b f(x(t),y(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$
$$\int_C f(x,y)\,dx = \int_a^b f(x(t),y(t))\,x'(t)\,dt$$
$$\int_C f(x,y)\,dy = \int_a^b f(x(t),y(t))\,y'(t)\,dt$$

Exercise 7.2.1. Evaluate the line integral, where C is the given curve

(a) $\displaystyle\int_C y^3\,ds$, $C : x = t^3,\ y = t, \quad 0 \le t \le 2$

(b) $\displaystyle\int_C xy\,ds$, $C : x = t^2,\ y = 2t, \quad 0 \le t \le 1$

(c) $\displaystyle\int_C xy^4\,ds$, C is the line segment from $(0,3)$ to $(4,6)$

(d) $\displaystyle\int_C (x^2y^3 - \sqrt{x})\,dy$, C is the arc of the curve $y = \sqrt{x}$ from $(1,1)$ to $(4,2)$

(e) $\displaystyle\int_C e^x\,dx$, C is the arc of the curve $x = y^3$ from $(-1,-1)$ to $(1,1)$

(f) $\displaystyle\int_C (x+2y)\,dx + x^2\,dy$, C consists of line segments from $(0,0)$ to $(2,1)$ and from $(2,1)$ to $(3,0)$

(g) $\displaystyle\int_C x^2\,dx + y^2\,dy$, C consists of the arc of the circle $x^2+y^2 = 4$ from $(2,0)$ to $(0,2)$ followed by the line segment from $(0,2)$ to $(4,3)$.

- We now suppose that C is a smooth space curve given by the parametric equations

$$x = x(t) \quad y = y(t) \quad z = z(t) \quad \text{where } a \le t \le b$$

or by a vector equation $\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j} + z(t)\,\mathbf{k}$. If f is a function of three variables that is continuous on some region containing C, then

$$\int_C f(x,y,z)\,ds = \int_a^b f(x(t),y(t),z(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt = \int_a^b f(\mathbf{r}(t))\,\left|\mathbf{r}'(t)\right|\,dt$$
$$\int_C f(x,y,z)\,dz = \int_a^b f(x(t),y(t),z(t))\,z'(t)\,dt$$

Exercise 7.2.2. Evaluate the line integral, where C is the given curve

(a) $\displaystyle\int_C xyz\,ds, \quad C: x = 2\sin t,\ y = t,\ z = -2\cos t, \quad 0 \le t \le \pi$

(b) $\displaystyle\int_C xyz^2\,ds$, C is the line segment from $(-1, 5, 0)$ to $(1, 6, 4)$

(c) $\displaystyle\int_C (x^2 + y^2 + z^2)\,ds$, where C: $x = t, y = \cos 2t, z = \sin 2t$, $0 \le t \le 2\pi$

(d) $\displaystyle\int_C xy\,e^{yz}\,dy$, where C: $x = t, y = t^2, z = t^3$, $0 \le t \le 1$

(e) $\displaystyle\int_C y\,dx + z\,dy + x\,dz$, where C: $x = \sqrt{t}, y = t, z = t^2$, $1 \le t \le 4$

(f) $\displaystyle\int_C z^2\,dx + x^2\,dy + y^2\,dz$, where C is the line segment from $(1, 0, 0)$ to $(4, 1, 2)$

Important formulas:

- Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C given by a vector function $\mathbf{r}(t)$, $a \le t \le b$. Then the **line integral of F along** C is
- $$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}'(t)\,dt = \int_C \mathbf{F}\cdot\mathbf{T}\,ds.$$
$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz \quad \text{where } \mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}.$$

Exercise 7.2.3. Evaluate the line integral $\int_C \boldsymbol{F}\cdot d\boldsymbol{r}$, where C is given by the vector function $\boldsymbol{r}(t)$

(a) $\boldsymbol{F}(x, y) = xy\boldsymbol{i} + 3y^2\boldsymbol{j}$ and $\boldsymbol{r}(t) = 11t^4\boldsymbol{i} + t^3\boldsymbol{j}$, $0 \le t \le 1$

(b) $\boldsymbol{F}(x, y, z) = (x + y)\boldsymbol{i} + (y - z)\boldsymbol{j} + z^2\boldsymbol{k}$ and $\boldsymbol{r}(t) = t^2\boldsymbol{i} + t^3\boldsymbol{j} + t^2\boldsymbol{k}$, $0 \le t \le 1$

(c) $\boldsymbol{F}(x, y, z) = \sin x\boldsymbol{i} + \cos y\boldsymbol{j} + xz\boldsymbol{k}$ and $\boldsymbol{r}(t) = t^3\boldsymbol{i} - t^2\boldsymbol{j} + t\boldsymbol{k}$, $0 \le t \le 1$

(d) $\boldsymbol{F}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + xy\boldsymbol{k}$ and $\boldsymbol{r}(t) = \cos t\boldsymbol{i} + \sin t\boldsymbol{j} + t\boldsymbol{k}$, $0 \le t \le \pi$

Exercise 7.2.4. Evaluate the line integral $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y) = e^{x-1}\,\mathbf{i} + xy\,\mathbf{j},$$

and C is given by

$$\mathbf{r}(t) = t^2\,\mathbf{i} + t^3\,\mathbf{j}, \quad 0 \le t \le 1.$$

Exercise 7.2.5. Find the work done by the force field $\boldsymbol{F}(x, y) = x^2\boldsymbol{i} + xy\boldsymbol{j}$ on a particle that moves once around the circle $x^2 + y^2 = 4$ oriented in the counter–clockwise direction.

Exercise 7.2.6. Find the work done by the force field

$$\mathbf{F}(x, y) = x\,\mathbf{i} + (y + 2)\,\mathbf{j}$$

in moving an object along one arch of the cycloid

$$\mathbf{r}(t) = (t - \sin t)\,\mathbf{i} + (1 - \cos t)\,\mathbf{j}, \qquad 0 \le t \le 2\pi.$$

Exercise 7.2.7. The force exerted by an electric charge at the origin on a charged particle at a point (x, y, z) with position vector

$$\mathbf{r} = \langle x, y, z \rangle$$

is

$$\mathbf{F}(\mathbf{r}) = \frac{K\mathbf{r}}{|\mathbf{r}|^3},$$

where K is a constant. Find the work done as the particle moves along a straight line from $(2, 0, 0)$ to $(2, 1, 5)$.

Exercise 7.2.8. (Mass and Center of Mass of a Wire)

(a) A thin wire is bent into the shape of a semicircle $x^2 + y^2 = 4$, $x \geq 0$. If the linear density is a constant k, find the mass and center of mass of the wire.

(b) A thin wire has the shape of the first-quadrant part of the circle with center the origin and radius a. If the density function is $\rho(x, y) = kxy$, find the mass and center of mass of the wire.

7.3 The Fundamental Theorem for Line Integrals

Important formulas:

- Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

- Theorem. $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in D.

- Theorem. Suppose $\mathbf{F}$ is a vector field that is continuous on an open connected region D. If $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D, then $\mathbf{F}$ is a conservative vector field on D; that is, there exists a function f such that $\nabla f = \mathbf{F}$.

- Theorem. If $\mathbf{F}(x, y) = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}$ is a conservative vector field, where P and Q have continuous first-order partial derivatives on a domain D, then throughout D we have

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

- Theorem. Let $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j}$ be a vector field on an open simply-connected region D. Suppose that P and Q have continuous first-order derivatives and

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \text{throughout } D$$

  Then $\mathbf{F}$ is conservative.

- Theorem. Let $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j}$ be a vector field on an open simply-connected region D. Suppose that P and Q have continuous first-order derivatives and

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \text{throughout } D$$

Then $\mathbf{F}$ is conservative.

Exercise 7.3.1. Determine whether or not $\boldsymbol{F}$ is a conservative vector field. If it is, find a function f such that $\boldsymbol{F} = \nabla f$.

(a) $\boldsymbol{F}(x,y) = (2x - 3y)\,\boldsymbol{i} + (-3x + 4y - 8)\,\boldsymbol{j}$

(b) $\boldsymbol{F}(x,y) = e^x \sin y\,\boldsymbol{i} + e^x \cos y\,\boldsymbol{j}$

(c) $\boldsymbol{F}(x,y) = e^x \cos y\,\boldsymbol{i} + e^x \sin y\,\boldsymbol{j}$

(d) $\boldsymbol{F}(x,y) = (3x^2 - 2y^2)\,\boldsymbol{i} + (4xy + 3)\,\boldsymbol{j}$

Exercise 7.3.2. Find a function f such that $\boldsymbol{F} = \nabla f$ and then use the function f to evaluate $\int_C \boldsymbol{F}\cdot d\boldsymbol{r}$ along a given curve C.

(a) $\boldsymbol{F}(x,y) = x^2\,\boldsymbol{i} + y^2\,\boldsymbol{j}$, and C is the arc of the parabola $y = 2x^2$ from $(-1,2)$ to $(2,8)$

(b) $\boldsymbol{F}(x,y) = xy^2\,\boldsymbol{i} + x^2y\,\boldsymbol{j}$, and C: $\boldsymbol{r}(t) = \big(t + \sin\frac{1}{2}\pi t, t + \cos\frac{1}{2}\pi t\big), \quad 0 \le t \le 1$

(c) $\boldsymbol{F}(x,y) = (1+xy)e^{xy}\,\boldsymbol{i} + x^2e^{xy}\,\boldsymbol{j}$, and C: $\boldsymbol{r}(t) = \cos t\,\boldsymbol{i} + 2\sin t\,\boldsymbol{j}, \quad 0 \le t \le \pi/2$

(d) $\boldsymbol{F}(x,y,z) = yz\,\boldsymbol{i} + xz\,\boldsymbol{j} + (xy + 2z)\,\boldsymbol{k}$, and C is the line segment from $(1,0,-2)$ to $(4,6,3)$

(e) $\boldsymbol{F}(x,y,z) = (y^2z + 2xz^2)\,\boldsymbol{i} + 2xyz\,\boldsymbol{j} + (xy^2 + 2x^2z)\,\boldsymbol{k}$, and C: $x = \sqrt{t},\ y = t+1,\ z = t^2,\ 0 \le t \le 1$.

Exercise 7.3.3. Show that the line integral is independent of path and evaluate the integral.

(a) $\displaystyle\int_C 2xe^{-y}\,dx + (2y - x^2e^{-y})\,dy, \quad C$ is any path from $(1,0)$ to $(2,1)$.

(b) $\displaystyle\int_C \sin y\,dx + (x\cos y - \sin y)\,dy, \quad C$ is any path from $(2,0)$ to $(1,\pi)$.

Exercise 7.3.4. Find the work done by the force field $\mathbf{F}$ in moving an object from P to Q.

(a) $\mathbf{F}(x,y) = 2y^{3/2}\,\mathbf{i} + 3x\sqrt{y}\,\mathbf{j}; \quad P(1,1),\ Q(2,4)$

(b) $\mathbf{F}(x,y) = e^{-y}\,\mathbf{i} - xe^{-y}\,\mathbf{j}; \quad P(0,1),\ Q(2,0)$.

7.4 Green's Theorem

Important formulas:

- **Green's Theorem** Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

- Suppose that the closed C is the boundary of the region D in the plane $\mathbb{R}^2$. We have

$$Area(D) = \oint_C x\,dy = -\oint_C y\,dx = \frac{1}{2}\oint_C x\,dy - y\,dx$$

Exercise 7.4.1. Evaluate the line integrals by two methods: directly and using Green's Theorem.

(a) $\oint_C (x-y)\,dx + (x+y)\,dy$, C is the circle with center the origin and radius 2.

(b) $\oint_C xy\,dx + x^2\,dy$, C is the rectangle with vertices $(0,0)$, $(3,0)$, $(3,1)$ and $(0,1)$.

(c) $\oint_C xy\,dx + x^2y^3\,dy$, C is the triangle with vertices $(0,0)$, $(1,0)$ and $(1,2)$.

(d) $\oint_C x^2y^2\,dx + xy\,dy$, C consists of the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ and the line segments from $(1,1)$ to $(0,1)$ and from $(0,1)$ to $(0,0)$.

Exercise 7.4.2. Use Green's Theorem to evaluate the line integral along the given positively oriented curve.

(a) $\oint_C xy^2\,dx + 2x^2y\,dy$, C is the tritangle with vertices $(0,0)$, $(2,2)$ and $(2,4)$.

(b) $\oint_C \cos y\,dx + x^2 \sin y\,dy$, C is the rectangle with vertices $(0,0)$, $(5,0)$, $(5,2)$ and $(0,2)$.

(c) $\oint_C (y + e^{\sqrt{x}})\,dx + (2x + \cos(y^2))\,dy$, C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

(d) $\oint_C y^4\,dx + 2xy^3\,dy$, C is the ellipse $x^2 + 2y^2 = 2$

(e) $\oint_C y^3\,dx - x^3\,dy$, C is the circle $x^2 + y^2 = 4$

Exercise 7.4.3. Use Green's Theorem to evaluate $\int_C \boldsymbol{F} \cdot d\boldsymbol{r}$.

(a) $\boldsymbol{F}(x,y) = (y\cos x - xy\sin x, xy + x\cos x)$ and C is the triangle from $(0,0)$ to $(0,4)$ to $(2,0)$ to $(0,0)$.

(b) $\boldsymbol{F}(x,y) = \left(e^{-x} + y^2, e^{-y} + x^2\right)$ and C consists of the arc of the curve $y = \cos x$ from $(-\pi/2, 0)$ to $(\pi/2, 0)$ and the line segment from $(\pi/2, 0)$ to $(-\pi/2, 0)$. is the triangle from $(0,0)$ to $(0,4)$ to $(2,0)$ to $(0,0)$.

(c) $\boldsymbol{F}(x,y) = (y - \cos y, x\sin y)$ and C is the circle $(x-3)^2 + (y+4)^2 = 4$ oriented clockwise.

(d) $\mathbf{F}(x,y) = \left\langle \sqrt{x^2+1},\ \tan^{-1}(x) \right\rangle$, where C is the triangle from $(0,0)$ to $(1,1)$ to $(0,1)$ to $(0,0)$.

7.5 Curl and Divergence

Important formulas:

- If $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P, Q, and R all exist, then the **curl of F** is the vector field on $\mathbb{R}^3$ defined by

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

- If $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\frac{\partial P}{\partial x}$, $\frac{\partial Q}{\partial y}$, and $\frac{\partial R}{\partial z}$ exist, then the **divergence of F** is the function of three variables defined by

$$\text{div } \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \nabla \cdot \mathbf{F}.$$

Exercise 7.5.1. Find curl and divergence of the vector field.

(a) $\boldsymbol{F}(x, y, z) = (x + yz)\boldsymbol{i} + (y + xz)\boldsymbol{j} + (z + xy)\boldsymbol{k}$

(b) $\boldsymbol{F}(x, y, z) = xy^2z^3\boldsymbol{i} + x^3yz^2\boldsymbol{j} + x^2y^3z\boldsymbol{k}$

(c) $\boldsymbol{F}(x, y, z) = xye^z\boldsymbol{i} + yze^x\boldsymbol{j}$

(d) $\boldsymbol{F}(x, y, z) = (e^x \sin y, e^y \sin x, e^z \sin x)$

Exercise 7.5.2. Determine whether or not the vector field $\boldsymbol{F}$ is conservative. If it is conservative, find a function f such that $\boldsymbol{F} = \nabla f$.

(a) $\boldsymbol{F}(x, y, z) = y^2z^3\boldsymbol{i} + 2xyz^3\boldsymbol{j} + 3xy^2z^2\boldsymbol{k}$

(b) $\boldsymbol{F}(x, y, z) = xyz^2\boldsymbol{i} + x^2yz^2\boldsymbol{j} + x^2y^2z\boldsymbol{k}$

(c) $\boldsymbol{F}(x, y, z) = \boldsymbol{i} + \sin z\boldsymbol{j} + y\cos z\boldsymbol{k}$

(d) $\boldsymbol{F}(x, y, z) = e^{yz}\boldsymbol{i} + xze^{yz}\boldsymbol{j} + xye^{yz}\boldsymbol{k}$

7.6 Parametric surfaces and their areas

Important formulas:

- If a smooth parametric surface $\mathcal{S}$ is given by the equation

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k} \qquad (u,v) \in D$$

  and $\mathcal{S}$ is covered just once as (u,v) ranges throughout the parameter domain D, then the **surface area** of $\mathcal{S}$ is

$$A(\mathcal{S}) = \iint_{\mathcal{S}} 1 dS = \iint_D |\mathbf{r}_u \times \mathbf{r}_v|\, dA$$

  where

$$\mathbf{r}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} + \frac{\partial z}{\partial u}\mathbf{k} \qquad \mathbf{r}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j} + \frac{\partial z}{\partial v}\mathbf{k}$$

- For the special case of a surface $\mathcal{S}$ with equation $z = f(x,y)$, where (x,y) lies in D and f has continuous partial derivatives, we have

$$A(\mathcal{S}) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$$

Exercise 7.6.1. Find the area of the surfaces

(a) The part of the plane $3x + 2y + z = 6$ that lies in the first octant.

(b) The part of the plane with vector equation $\boldsymbol{r}(u,v) = (u+v, 2-3u, 1+u-v)$ that is given by $0 \le u \le 2$, $-1 \le v \le 1$.

(c) The part of the plane $x + 2y + 3z = 1$ that lies inside the cylinder $x^2 + y^2 = 3$.

(d) The part of the cone $z = \sqrt{x^2+y^2}$ that lies between the plane $y = x$ and the cylinder $y = x^2$.

(e) The surface $z = \frac{2}{3}\left(x^{3/2} + y^{3/2}\right)$, $0 \le x \le 1$, $0 \le y \le 1$.

(f) The part of the surface $z = 1 + 3x + 2y^2$ that lies above the triangle with vertices $(0,0)$, $(0,1)$, and $(2,1)$.

(g) The part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.

(h) The part of the paraboloid $x = y^2 + z^2$ that lies inside the cylinder $y^2 + z^2 = 9$.

(i) The part of the surface $y = 4x + z^2$ that lies between the planes $x = 0$, $x = 1$, $z = 0$, and $z = 1$.

(j) The helicoid (or spiral ramp) with vector equation

$$\boldsymbol{r}(u,v) = u\,\cos v\,\boldsymbol{i} + u\,\sin v\,\boldsymbol{j} + v\,\boldsymbol{k}, \quad 0 \le u \le 1, 0 \le v \le \pi.$$

(k) The surface with parametric equations $x = u^2$, $y = uv$, $z = \frac{1}{2}v^2$, $0 \le u \le 1$, $0 \le v \le 2$.

(l) The part of the sphere $x^2 + y^2 + z^2 = b^2$ that lies inside the cylinder $x^2 + y^2 =^2$, where $0 < a < b$.

7.7 Surface Integrals

Surface integrals of functions

Important formulas:

- Suppose that a surface S has a vector equation

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k}, \quad (u,v) \in D,$$

$$\iint_S f(x,y,z)\,dS = \iint_D f(\mathbf{r}(u,v))\left|\mathbf{r}_u \times \mathbf{r}_v\right| dA.$$

- Suppose that the surface $\mathcal{S}$ is defined by equation $z = g(x,y)$, $(x,y) \in D \subset \mathbb{R}^2$. Then

$$\iint_S f(x,y,z)\,dS = \iint_D f(x,y,g(x,y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\,dA$$

Exercise 7.7.1. Evaluate the surface integrals.

(a) $\iint_S (x+y+z)\,dS$, where S is the parallelogram with parametric equations

$$x = u+v,\ y = u-v,\ z = 1+2u+v, \quad 0 \le u \le 2,\ 0 \le v \le 1.$$

(b) $\iint_S xyz\,dS$, where S is the cone with parametric equations

$$x = u\cos v,\ y = u\sin v,\ z = u, \quad 0 \le u \le 1,\ 0 \le v \le \pi/2.$$

(c) $\iint_S y\,dS$, where S is the helicoid with vector equation

$$\boldsymbol{r}(u,v) = (u\cos v, u\sin v, v), \quad 0 \le u \le 1, 0 \le v \le \pi.$$

(d) $\iint_S (x^2+y^2)\,dS$, where S is the surface with vector equation

$$\boldsymbol{r}(u,v) = \left(2uv, u^2-v^2, u^2+v^2\right), \quad u^2+v^2 \le 1.$$

(e) $\iint_S x^2yz\,dS$, where S is the part of the plane $z = 1+2x+3y$ that lies above the rectangle $[0,3]\times[0,2]$.

(f) $\iint_S xz\,dS$, where S is the part of the plane $2x+2y+z = 4$ that lies in the first octant.

(g) $\iint_S x\,dS$, where S is the triangular region with vertices $(1,0,0)$, $(0,-2,0)$, and $(0,0,4)$.

(h) $\iint_S y\,dS$, where S is the surface $z = \frac{2}{3}(x^{3/2}+y^{3/2})$, $0 \le x \le 1$, $0 \le y \le 1$.

(i) $\iint_S x^2z\,dS$, where S is the part of the cone $z^2 = x^2+y^2$ that lies between the planes $z = 1$ and $z = 3$.

(j) $\iint_S z\,dS$, where S is the surface $x = y + 2z^2,\ 0 \le y \le 1,\ 0 \le z \le 1$.

(k) $\iint_S y\,dS$, where S is the part of the paraboloid $y = x^2+z^2$ that lies inside the cylinder $x^2+z^2 = 4$.

(l) $\iint_S y^2\,dS$, where S is the part of the sphere $x^2+y^2+z^2 = 4$ that lies inside the cylinder $x^2+y^2 = 1$ and above the xy-plane.

(m) $\iint_S (x^2z + y^2z)\,dS$, where S is the hemisphere $x^2 + y^2 + z^2 = 4,\ z \ge 0$.

(n) $\iint_S xz\,dS$, where S is the boundary of the region enclosed by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0$ and $x + y = 5$.

(o) $\iint_S (z+x^2y)\,dS$, where S is the part of the cylinder $y^2+z^2 = 1$ that lies between the planes $x = 0$ and $x = 3$ in the first octant.

(p) $\iint_S (x^2 + y^2 + z^2)\,dS$, where S is the part of the cylinder $x^2 + y^2 = 9$ between the planes $z = 0$ and $z = 2$, together with its top and bottom disks.

Surface integrals of vector fields

Important formulas:

- If $\mathbf{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\mathbf{n}$, then the **surface integral of F over** S is

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\mathbf{n}\,dS.$$

  This integral is also called the **flux** of $\mathbf{F}$ across S.

- If S is given by a vector function $\mathbf{r}(u,v)$, $(u,v) \in D \subset \mathbb{R}^2$, then

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_D \mathbf{F}\cdot(\mathbf{r}_u \times \mathbf{r}_v)\ dA.$$

- In the case of a surface S given by a graph $z = g(x,y)$, $(x,y) \in D \subset \mathbb{R}^2$ with upward direction then

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_D \left(-P\frac{\partial g}{\partial x} - Q\frac{\partial g}{\partial y} + R\right)\ dA.$$

Exercise 7.7.2. Evaluate the surface integral $\iint_S \boldsymbol{F}\cdot d\boldsymbol{S}$ for the given vector field $\boldsymbol{F}$ and the oriented surface S. In other words, find the flux of $\boldsymbol{F}$ across S. For closed surfaces, use the positive (outward) orientation.

(a) $\boldsymbol{F}(x,y,z) = ze^{xy}\,\boldsymbol{i} - 3ze^{xy}\,\boldsymbol{j} + xy\,\boldsymbol{k}$, where S is the parallelogram with parametric equations $x = u + v$, $y = u - v$, $z = 1 + 2u + v$, $\quad 0 \le u \le 2$, $0 \le v \le 1$ with upward orientation.

(b) $\boldsymbol{F}(x,y,z) = z\,\boldsymbol{i} + y\,\boldsymbol{j} + x\,\boldsymbol{k}$, where S is the helicoid with vector equation

$$\boldsymbol{r}(u,v) = (u\,\cos v, u\,\sin v, v)\,, \quad 0 \le u \le 1, 0 \le v \le \pi.$$

with upward orientation.

(c) $\boldsymbol{F}(x,y,z) = xy\,\boldsymbol{i} + yz\,\boldsymbol{j} + zx\,\boldsymbol{k}$, where S is the part of the para boloid $z = 4 - x^2 - y^2$ that lies above the square $0 \le x \le 1$, $0 \le y \le 1$ and has upward orientation.

(d) $\boldsymbol{F}(x,y,z) = -x\,\boldsymbol{i} - y\,\boldsymbol{j} + z^3\,\boldsymbol{k}$, where S is the part of the cone $z = \sqrt{x^2+y^2}$ between the planes $z = 1$ and $z = 3$ with downward orientation.

(e) $\mathbf{F}(x,y,z) = x\mathbf{i} - z\mathbf{j} + y\mathbf{k}$, where S is the part of the sphere $x^2 + y^2 + z^2 = 4$ in the first octant, with orientation toward the origin.

(f) $\mathbf{F}(x,y,z) = xz\,\mathbf{i} + x\mathbf{j} + y\mathbf{k}$, where S is the hemisphere $x^2 + y^2 + z^2 = 25,\ y \ge 0$, oriented in the direction of the positive y-axis.

(g) $\mathbf{F}(x,y,z) = y\mathbf{j} - z\mathbf{k}$, where S consists of the paraboloid $y = x^2 + z^2,\ 0 \le y \le 1$, and the disk $x^2 + z^2 \le 1,\ y = 1$.

(h) $\mathbf{F}(x,y,z) = xy\mathbf{i} + 4x^2\mathbf{j} + yz\mathbf{k}$, where S is the surface $z = xe^y,\ 0 \le x \le 1,\ 0 \le y \le 1$, with upward orientation.

(i) $\mathbf{F}(x,y,z) = x\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}$, where S is the cube with vertices $(\pm1, \pm1, \pm1)$.

(j) $\mathbf{F}(x,y,z) = x\mathbf{i} + y\mathbf{j} + 5\mathbf{k}$, where S is the boundary of the region enclosed by the cylinder $x^2 + z^2 = 1$ and the planes $y = 0$ and $x + y = 2$.

(k) $\mathbf{F}(x,y,z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$, where S is the boundary of the solid half-cylinder $0 \le z \le \sqrt{1-y^2},\ 0 \le x \le 2$.

(l) $\mathbf{F}(x,y,z) = y\mathbf{i} + (z-y)\mathbf{j} + x\mathbf{k}$, where S is the surface of the tetrahedron with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Exercise 7.7.3. Find the flux of

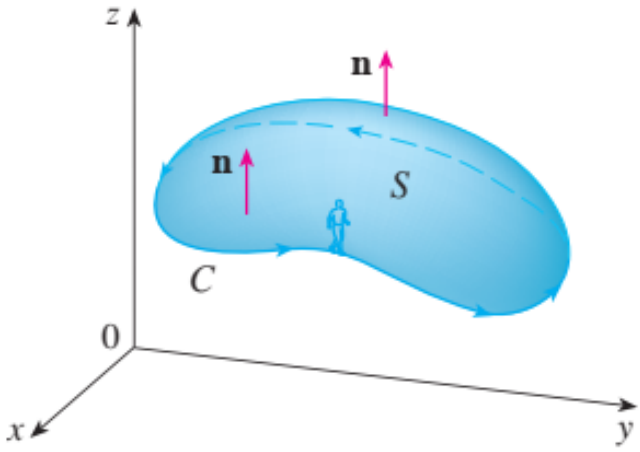
$$\mathbf{F}(x,y,z) = \sin(xyz)\,\mathbf{i} + x^2y\,\mathbf{j} + z^2e^{x/5}\,\mathbf{k}$$

across the part of the cylinder $4y^2 + z^2 = 4$ that lies above the xy-plane and between the planes $x = -2$ and $x = 2$ with upward orientation.

7.8 Stokes' Theorem

Stokes' Theorem: Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S}.$$



Exercise 7.8.1. Use Stokes' Theorem to evaluate $\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}$.

(a) $\mathbf{F}(x, y, z) = 2y \cos z\, \mathbf{i} + e^x \sin z\, \mathbf{j} + xe^y\, \mathbf{k}$, and

S is the hemisphere $x^2 + y^2 + z^2 = 9,\ z \geq 0$, oriented upward.

(b) $\mathbf{F}(x, y, z) = x^2 z^2\, \mathbf{i} + y^2 z^2\, \mathbf{j} + xyz\, \mathbf{k}$, and

S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$, oriented upward.

(c) $\mathbf{F}(x, y, z) = \tan^{-1}(x^2 yz^2)\, \mathbf{i} + x^2 y\, \mathbf{j} + x^2 z^2\, \mathbf{k}$, and

S is the cone $x = \sqrt{y^2 + z^2},\ 0 \leq x \leq 2$, oriented in the direction of the positive x-axis.

(d) $\mathbf{F}(x, y, z) = xyz\, \mathbf{i} + xy\, \mathbf{j} + x^2 yz\, \mathbf{k}$, and

S consists of the top and the four sides (but not the bottom) of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, oriented outward.

(e) $\mathbf{F}(x, y, z) = e^{xy}\, \mathbf{i} + e^x\, \mathbf{j} + x^2\, \mathbf{k}$, and

S is the half of the ellipsoid $4x^2 + y^2 + 4z^2 = 4$ that lies to the right of the xz-plane, oriented in the direction of the positive y-axis.

Exercise 7.8.2. Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$. In each case C is oriented counterclockwise as viewed from above.

(a) $\mathbf{F}(x, y, z) = (x + y^2)\mathbf{i} + (y + z^2)\mathbf{j} + (z + x^2)\mathbf{k}$, and

C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

(b) $\mathbf{F}(x, y, z) = \mathbf{i} + (x + yz)\mathbf{j} + (xy - \sqrt{z})\mathbf{k}$, and

C is the boundary of the part of the plane $3x + 2y + z = 1$ in the first octant.

(c) $\mathbf{F}(x, y, z) = yz\, \mathbf{i} + 2xz\, \mathbf{j} + e^{xy}\mathbf{k}$, and

C is the circle $x^2 + y^2 = 16,\ z = 5$.

(d) $\mathbf{F}(x, y, z) = xy\,\mathbf{i} + 2z\,\mathbf{j} + 3y\,\mathbf{k}$, and

C is the curve of intersection of the plane $x + z = 5$ and the cylinder $x^2 + y^2 = 9$.

Exercise 7.8.3. Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y, z) = x^2 z\,\mathbf{i} + xy^2\,\mathbf{j} + z^2\,\mathbf{k},$$

and C is the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 9$, oriented counterclockwise as viewed from above.

Exercise 7.8.4. Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y, z) = x^2 y\,\mathbf{i} + \tfrac{1}{3}x^3\,\mathbf{j} + xy\,\mathbf{k},$$

and C is the curve of intersection of the hyperbolic paraboloid $z = y^2 - x^2$ and the cylinder $x^2 + y^2 = 1$, oriented counterclockwise as viewed from above.

7.9 The Divergence Theorem

The Divergence Theorem: Let E be a simple solid region and let S be the boundary surface of E, given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F}\, dV.$$

Exercise 7.9.1. Verify that the Divergence Theorem is true for the vector field $\mathbf{F}$ on the region E.

(a) $\mathbf{F}(x, y, z) = 3x\,\mathbf{i} + xy\,\mathbf{j} + 2xz\,\mathbf{k}$, and

E is the cube bounded by the planes $x = 0,\ x = 1,\ y = 0,\ y = 1,\ z = 0,\ z = 1$.

(b) $\mathbf{F}(x, y, z) = x^2\,\mathbf{i} + xy\,\mathbf{j} + z\,\mathbf{k}$, and

E is the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy-plane.

(c) $\mathbf{F}(x, y, z) = \langle z, y, x\rangle$, and E is the solid ball $x^2 + y^2 + z^2 \le 16$.

(d) $\mathbf{F}(x, y, z) = \langle x^2, -y, z\rangle$, and E is the solid cylinder $y^2 + z^2 \le 9,\ 0 \le x \le 2$.

(e) $\mathbf{F}(x, y, z) = 3xy^2\,\mathbf{i} + xe^z\,\mathbf{j} + z^3\,\mathbf{k}$, and

S is the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and the planes $x = -1$ and $x = 2$.

(f) $\mathbf{F}(x, y, z) = (x^3 + y^3)\,\mathbf{i} + (y^3 + z^3)\,\mathbf{j} + (z^3 + x^3)\,\mathbf{k}$, and

S is the sphere with center the origin and radius 2.

(g) $\mathbf{F}(x, y, z) = x^2 \sin y\,\mathbf{i} + x \cos y\,\mathbf{j} - xz \sin y\,\mathbf{k}$, and

S is the "fat sphere" $x^8 + y^8 + z^8 = 8$.

(h) $\mathbf{F}(x, y, z) = z\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$, and

S is the surface of the tetrahedron enclosed by the coordinate planes and the plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1,$$

where a, b, c are positive numbers.

(i) $\mathbf{F}(x, y, z) = (\cos z + xy^2)\,\mathbf{i} + xe^z\,\mathbf{j} + (\sin y + x^2 z)\,\mathbf{k}$, and

S is the surface of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

(j) $\mathbf{F}(x, y, z) = x^4\,\mathbf{i} - x^2 z^3\,\mathbf{j} + 4xy^2 z\,\mathbf{k}$, and

S is the surface of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = x + 2$ and $z = 0$.

(k) $\mathbf{F} = |\mathbf{r}|\,\mathbf{r}$, where $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$, and

S consists of the hemisphere $z = \sqrt{1 - x^2 - y^2}$ and the disk $x^2 + y^2 \leq 1$ in the xy-plane.

Chapter 8

Sequences and Series

8.1 Sequences

Exercise 8.1.1. Does the sequence appear to have a limit? If so, calculate it. If not, explain why.

(a) $a_n = \dfrac{3n}{1+6n}$ (b) $a_n = 1 + \left(\dfrac{-1}{2}\right)^n$ (c) $a_n = 2 + \left(\dfrac{(-1)^n}{n}\right)$ (d) $a_n = 1 + \left(\dfrac{10^n}{9^n}\right)$.

Exercise 8.1.2. Determine whether the sequence converges or diverges. If it converges, find the limit.

(a) $a_n = \dfrac{n^3}{n^3+1}$

(b) $a_n = e^{1/n}$

(c) $a_n = \dfrac{(-1)^{n+1}n}{n+\sqrt{n}}$

(d) $a_n = \cos(n/2)$

(e) $a_n = \dfrac{e^n + e^{-n}}{e^{2n}-1}$

(f) $a_n = \dfrac{\ln n}{\ln 2n}$

(g) $a_n = n^2 e^{-n}$

(h) $a_n = \dfrac{\cos^2 n}{2^n}$

(i) $a_n = \sqrt[n]{2^{1+3n}}$

(j) $a_n = \left(1 + \dfrac{2}{n}\right)^n$

(k) $a_n = \ln(2n^2+1) - \ln(n^2+1)$

(l) $a_n = n - \sqrt{n+1}\sqrt{n+3}$

(m) $a_n = \dfrac{n!}{2^n}$

(n) $a_n = \dfrac{(-3)^n}{n!}$

Exercise 8.1.3. Let $a_n = \dfrac{n}{5n+2}$.

(a) Determine whether $\{a_n\}$ is convergent.

(b) Determine whether $\sum_{n=1}^{\infty} a_n$ is convergent.

Exercise 8.1.4. Similar questions as in Exercise 8.1.3 with:

(a) $a_n = \dfrac{1}{n(n+2)}$ (b) $a_n = \dfrac{1}{n(n+5)}$

8.2 Series

Exercise 8.2.1. Determine whether the series is convergent or divergent. If it is convergent, find its sum

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{2}{n^2-1}$ (b) $\displaystyle\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$ (c) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n(n+3)}$

Exercise 8.2.2. Determine whether the *geometric* series is convergent or divergent. If it is convergent, find its sum.

(a) $3+2+\frac{4}{3}+\frac{8}{9}+\dots$

(b) $\frac{1}{8}-\frac{1}{4}+\frac{1}{2}-1+\dots$

(c) $\displaystyle\sum_{n=1}^{\infty} 6(0.9)^{n-1}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{10^n}{(-9)^{n-1}}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{(-3)^{n-1}}{4^n}$

(f) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(\sqrt{2})^n}$

(g) $\displaystyle\sum_{n=0}^{\infty} \frac{\pi^n}{3^{n+1}}$

(h) $\displaystyle\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}}$

Exercise 8.2.3. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2n}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{2n-3}$

(c) $\displaystyle\sum_{k=2}^{\infty} \frac{k^2}{k^2-1}$

(d) $\displaystyle\sum_{k=1}^{\infty} \frac{k(k+2)}{(k+3)^2}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{1+2^n}{3^n}$

(f) $\displaystyle\sum_{n=1}^{\infty} \frac{1+3^n}{2^n}$

(g) $\displaystyle\sum_{n=1}^{\infty} \sqrt[n]{2}$

(h) $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n^2+1}{2n^2+1}\right)$

(i) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{3}{7^n}+\frac{2}{n}\right)$

(j) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{e^n}+\frac{1}{n(n+1)}\right)$

(k) $\displaystyle\sum_{n=1}^{\infty} \frac{e^n}{n^2}$

Exercise 8.2.4. Determine whether the series converges or diverges. (Hint: you may use the Ratio Test or the Root Test)

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{a^n}$ $(a>0)$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{n+2^n}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{(2+\frac{1}{n})^n}$

(f) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2-5n+1}{n^2-4n+2}\right)^{n^2}$

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

(h) $\displaystyle\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{100^n}$

(j) $\displaystyle\sum_{n=1}^{\infty} \frac{3-\cos n}{n^{2/3}-2}$

(k) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+1}\right)^n$

(l) $\displaystyle\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$

(m) $\displaystyle\sum_{n=2}^{\infty} \frac{n}{(\ln n)^n}$

(n) $\displaystyle\sum_{n=1}^{\infty} \frac{2\cdot 4\cdot 6\cdot \ldots \cdot (2n)}{n!}$

Exercise 8.2.5. Determine whether the series converges or diverges.

(a) $\displaystyle\sum_{n=1}^{\infty} ne^{-n}$

(b) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{2n+1}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n^3}$

(d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

(e) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

(f) $\displaystyle\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{e^n}$

Exercise 8.2.6. Find the values of p for which the series is convergent. (Hint: you may use the Integral Test)

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$ (b) $\displaystyle\sum_{n=3}^{\infty} \frac{1}{n \ln n[\ln(\ln n)]^p}$ (c) $\displaystyle\sum_{n=1}^{\infty} n(1+n^2)^p$ (d) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n^p}$

Exercise 8.2.7. Determine whether the series converges or diverges. (Hint: you may use the Comparison Test)

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2n^3+1}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^3}{n^4+1}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n\sqrt{n}}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{n-1}{n^2\sqrt{n}}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$

(f) $\displaystyle\sum_{n=1}^{\infty} \frac{4+3^n}{2^n}$

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^2+1}$

(h) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2-1}{3n^4+1}$

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{n-1}{n4^n}$

(j) $\displaystyle\sum_{n=1}^{\infty} \frac{1+\sin n}{10^n}$

(k) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n-1}$

(l) $\displaystyle\sum_{n=1}^{\infty} \frac{2+(-1)^n}{n\sqrt{n}}$

(m) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$

(n) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$

(o) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2n+3}$

(p) $\displaystyle\sum_{n=1}^{\infty} \frac{1+4^n}{1+3^n}$

(q) $\displaystyle\sum_{n=1}^{\infty} \frac{n+4^n}{n+6^n}$

(r) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n+2}}{2n^2+n+1}$

(s) $\displaystyle\sum_{n=1}^{\infty} \frac{n+2}{(n+1)^3}$

(t) $\displaystyle\sum_{n=1}^{\infty} \frac{e^{1/n}}{n}$

(u) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!}$

(v) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(w) $\displaystyle\sum_{n=1}^{\infty} \sin\frac{1}{n}$

Exercise 8.2.8. (a) $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$

(b) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1}\frac{3n+1}{n(n+1)}$

(c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\frac{3n-1}{2n+1}$

(d) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\frac{n}{\sqrt{n^3+2}}$

(e) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\frac{n}{10^n}$

(f) $\displaystyle\sum_{n=2}^{\infty} (-1)^n\frac{n}{\ln n}$

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{\cos n\pi}{n^{3/4}}$

(h) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\sin\frac{\pi}{n}$

(i) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\cos\frac{\pi}{n}$

Exercise 8.2.9. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$

(b) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1}\frac{2^n}{n^4}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt[4]{n}}$

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{100^n}$

(f) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\frac{n}{\sqrt{n^3+2}}$

8.3 Power series

Exercise 8.3.1. Find the values of x for which the series converges. Find the sum of the series for those values of x.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{3^n}$

(b) $\displaystyle\sum_{n=1}^{\infty} (x-4)^n$

(c) $\displaystyle\sum_{n=0}^{\infty} 4^n x^n$

(d) $\displaystyle\sum_{n=0}^{\infty} \frac{(x+3)^n}{2^n}$

(e) $\displaystyle\sum_{n=0}^{\infty} \frac{\cos^n x}{2^n}$

Exercise 8.3.2. Find the radius of convergence and interval of convergence of the series.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n+1}$

(c) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n^3}$

(d) $\displaystyle\sum_{n=1}^{\infty} \sqrt{n}x^n$

(e) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n!}$

(f) $\displaystyle\sum_{n=1}^{\infty} n^n x^n$

(g) $\displaystyle\sum_{n=1}^{\infty} (-1)^n\frac{n^2 x^n}{2^n}$

(h) $\displaystyle\sum_{n=1}^{\infty} \frac{10^n x^n}{n^3}$

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-2)^n x^n}{\sqrt[4]{n}}$

(j) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{5^n n^5}$

(k) $\displaystyle\sum_{n=2}^{\infty} (-1)^n\frac{x^n}{4^n \ln n}$

(l) $\displaystyle\sum_{n=0}^{\infty} (-1)^n\frac{x^{2n}}{(2n)!}$

(m) $\displaystyle\sum_{n=0}^{\infty} \frac{(x-2)^n}{n^n}$

(n) $\displaystyle\sum_{n=0}^{\infty} (-1)^n\frac{(x-3)^n}{2n+1}$

(o) $\displaystyle\sum_{n=1}^{\infty} \frac{3^n(x+4)^n}{\sqrt{n}}$

(p) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{4^n}(x+1)^n$

(q) $\displaystyle\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^n}$

Exercise 8.3.3. Find a power series representation for $\ln(1-x)$ and its radius of convergence.

Exercise 8.3.4. Find a power series representation for $f(x) = \dfrac{x^2}{1-x^2}$

Exercise 8.3.5. Find the Maclaurin series for $\cos x$

Exercise 8.3.6. Find the Maclaurin series for $x\cos x$

Exercise 8.3.7. Find the Maclaurin series for

(a) $\dfrac{1}{1-x}$

(b) xe^x

(c) $\ln(1+x)$

(d) $\cos 3x$

(e) $e^x + e^{3x}$

(f) $\sin^2 x$

(g) $\cos x^2$

(h) $\ln(1+x^2)$

Exercise 8.3.8. Use the Maclaurin series for e^x to calculate $e^{-0.2}$ correct to five decimal places.

Exercise 8.3.9. Find the Maclaurin series of $\tan x$.

Contents

1 Integration **2**

1.1 Integration techniques 2

1.2 Fundamental theorems of Calculus 3

1.3 Improper integrals 4

2 Applications of Integration **6**

2.1 Average value of a function, Areas, volumes, arc lengths 6

2.2 Others 8

3 Differential Equations **9**

3.1 Separable Equations 10

3.2 Linear Equations 11

3.3 Second-order Linear Differential Equations 12

4 Vector and geometry of space **14**

4.1 Vectors 14

4.2 Equations of Lines and Planes 15

4.3 Three-Dimensional Coordinate Systems 17

5 Partial Derivatives **19**

5.1 Limits 19

5.2 Continuity 20

5.3 Partial derivatives 20

5.4 Local and absolute Extremums 22

6 Multiple Integrals **24**

6.1 Double Integrals 24

6.2 Polar coordinates - Double integrals 25

6.3 Triple Integrals 26

6.4 Cylindrical coordinates - Triple integrals 27

6.5 Spherical coordinates 27

7 Vector Calculus **29**
7.1 Vector Fields 29
7.2 Line Integrals 30
7.3 The Fundamental Theorem for Line Integrals 32
7.4 Green's Theorem 34
7.5 Curl and Divergence 35
7.6 Parametric surfaces and their areas 36
7.7 Surface Integrals 37
7.8 Stokes' Theorem 39
7.9 The Divergence Theorem 41

8 Sequences and Series **43**
8.1 Sequences 43
8.2 Series 43
8.3 Power series 46